ProcessTensors.jl: An ITensor-native framework for non-Markovian open quantum dynamics

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Abstract

We present ProcessTensors.jl, a pedagogical Julia package for understanding, constructing, and using process tensors with ITensor. The paper introduces process tensors from the ground up and connects the equations to tensor diagrams, tagged ITensor indices, and working code. Through guided examples, users can build process tensors, apply different sequences of quantum instruments, and use testers with ancillary systems. We implement exact and Automated Compression of Environments (ACE) constructions and benchmark them against established software. We also review the current process tensor software landscape and outline extensions through improved ACE compression, driven and dissipative bath modes, Gaussian PT-TEMPO, and chain-mapped bath constructions. Our goal is to make process tensors approachable for new users while providing a clear foundation for future development of this package.

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1 Introduction

In principle, describing an open quantum system is easy: write down a giant wavefunction for the entire universe, place it in an even more gigantic Hilbert space, and evolve it according to the Schrödinger equation. Somewhere inside that wavefunction lies everything one could wish to know about the system of interest. In practice, of course, we cannot keep track of

every surrounding particle, lattice vibration, stray electromagnetic field, or experimental imperfection and calculate their joint correlated dynamics. We therefore do what physicists have always done when the universe becomes inconveniently large: draw a boundary. The degrees of freedom we wish to study are called the *system*; everything beyond this boundary is called the *environment* or *bath*. Open quantum-system theory asks how the bath affects the system without requiring us to simulate the bath in all its microscopic detail.

The most familiar answers we would see in this regard are those that rely on weak-coupling, short-memory, or related approximations on the baths that lead to time-local master equations. In the Markovian limit, the system does not need a record of its earlier history in order to determine what happens next. Markovian master equations provide an extremely successful workhorse for open quantum dynamics, with the Gorini–Kossakowski–Sudarshan–Lindblad form supplying the standard generator of quantum dynamical semigroups [1, 2]. Nevertheless, weak coupling and a clean separation of timescales are not available in every problem. Structured reservoirs, strong system–bath coupling, slow environmental modes, and delayed feedback can all preserve information about earlier dynamics [3,4]. In these regimes, memory is part of the physics rather than a correction that can safely be ignored.

The difficulty is not merely that the environment remembers, but that the system may give us no indication of *what* it remembers if we trace out the bath. To see this, consider two joint system–environment states at some time t_1,

$$\rho_{SE}^{(1)}(t_1) \qquad \text{and} \qquad \rho_{SE}^{(2)}(t_1), \tag{1}$$

which reduce to exactly the same system state,

$$\mathrm{Tr}_E\Big[\rho_{SE}^{(1)}(t_1)\Big] = \mathrm{Tr}_E\Big[\rho_{SE}^{(2)}(t_1)\Big] = \rho_S(t_1). \tag{2}$$

As far as the system density matrix is concerned, these two situations are indistinguishable. The environment, however, may disagree. It can retain different states or different correlations with the system in the two cases. Consequently, evolving both joint states with the same system–environment unitary need not produce the same reduced state at a later time:

$$\mathrm{Tr}_E\Big[U_{2:1}\rho_{SE}^{(1)}(t_1)U_{2:1}^\dagger\Big] \neq \mathrm{Tr}_E\Big[U_{2:1}\rho_{SE}^{(2)}(t_1)U_{2:1}^\dagger\Big]. \tag{3}$$

Thus, even perfect knowledge of the present system state need not be sufficient to predict its future. The information distinguishing the two experiments has simply moved somewhere beyond the boundary we drew earlier.

The limitation becomes sharper when we intervene in our system along the way. Imagine performing two experiments with different histories and, at an intermediate time t_1, discarding the system state and preparing the same fresh state P in both. This operation—often called a *causal break*—removes any information that could pass directly from the past to the future through the system. Immediately afterward, both experiments therefore contain the same reduced state,

$$\rho_S^{(1)}(t_1^+) = \rho_S^{(2)}(t_1^+) = P. \tag{4}$$

Nevertheless, the intervention acts on a system that may already be correlated with its environment. A measurement outcome, for example, can condition the environment into a corresponding state, while repreparing P makes the system look identical in every run. If the later dynamics still depends on the earlier preparation or measurement outcome, then the memory must have travelled across the causal break through the environment. Such memory can remain hidden in an uninterrupted reduced trajectory, and can even occur when familiar two-time criteria describe the unperturbed evolution as Markovian [5].

This leaves a state-to-state description with an awkward question. What map should take P at t_1 to the state at t_2, if the same P can lead to different futures depending on what happened

before it was prepared? There is no contradiction in quantum mechanics here; rather, we have asked the reduced state to remember information that it no longer contains. The missing input is the history of interventions—the preparations, controls, and measurements through which the system and its environment arrived at the present experiment. A complete description of the quantum process must therefore accept an entire ordered sequence of operations, rather than only a density matrix at one particular time.

The process tensor provides precisely such a description [6–8]. For a sequence of interventions $(\mathcal{A}_0, \mathcal{A}_1, \ldots, \mathcal{A}_{n-1})$, it returns the final system state,

$$\rho_S(t_n) = \mathcal{T}_{n:0}\big[\mathcal{A}_{n-1}, \ldots, \mathcal{A}_0\big], \tag{5}$$

or, when the interventions contain measurements, the corresponding outcome probabilities and conditional states. A reduced trajectory tells us what happened under one prescribed history; a process tensor tells us what could have happened had we intervened differently along the way. It does not reveal every microscopic bath degree of freedom, but retains their complete influence on experiments performed through the system (the mathematical construction behind this statement—including its Choi representation and temporal input and output spaces—will be developed carefully in Sec. 3). For the present discussion, its operational meaning is enough. This makes causal breaks, multi-time correlations, instrument sequences, Markov order, and correlations across temporal partitions accessible within one description. In the Markovian limit, the process separates into independent transformations between successive times; otherwise, its temporal parts remain correlated, recording how information stored in the environment can return later. Memory thus becomes part of the object being manipulated rather than a feature inferred from a single trajectory.

The generality of a process tensor comes at an exponential cost: every additional intervention time introduces another pair of system indices, thus causing the exponential growth of the representation space of a process tensor, similar to how a many-body wavefunction space grows with the number of particles. Tensor networks make this manageable by representing the process as a matrix product operator in time, or PT-MPO [9]. The open legs provide the slots for system operations, while the internal bonds carry correlations between different parts of the process. When these temporal correlations are compressible, singular-value truncations reduce the representation to a controllable bond dimension. The process tensor is therefore not made small by assumption; it is made useful whenever the environment stores its relevant memory in a compressible temporal space.

1.1 From microscopic baths to process tensors

Having a place to store the memory is only half of the problem. We still need to work out what the environment writes into it. For simulations based on a microscopic bath model, constructing a sufficiently accurate and compact PT-MPO is often the main computational task. Once it is available, changing a preparation, inserting a measurement, or calculating a multi-time correlation amounts to connecting different system operations to its open legs. For a fixed bath model, coupling, initial bath state, and time grid, the same environmental tensor can therefore support many different experiments without repeating its construction.

Figure 1 organises several established routes to this object according to the information used to describe the bath: its Gaussian correlations, its individual microscopic modes, or its spatial network. These are overlapping representations rather than disjoint classes of physical environments. A collection of harmonic oscillators, for example, may be integrated out analytically, treated mode by mode, or transformed into a chain. Likewise, a discrete bath need not be non-Gaussian, and a continuous spectrum does not by itself imply Gaussian statistics. The useful question is which structure we can exploit for what practical purposes.

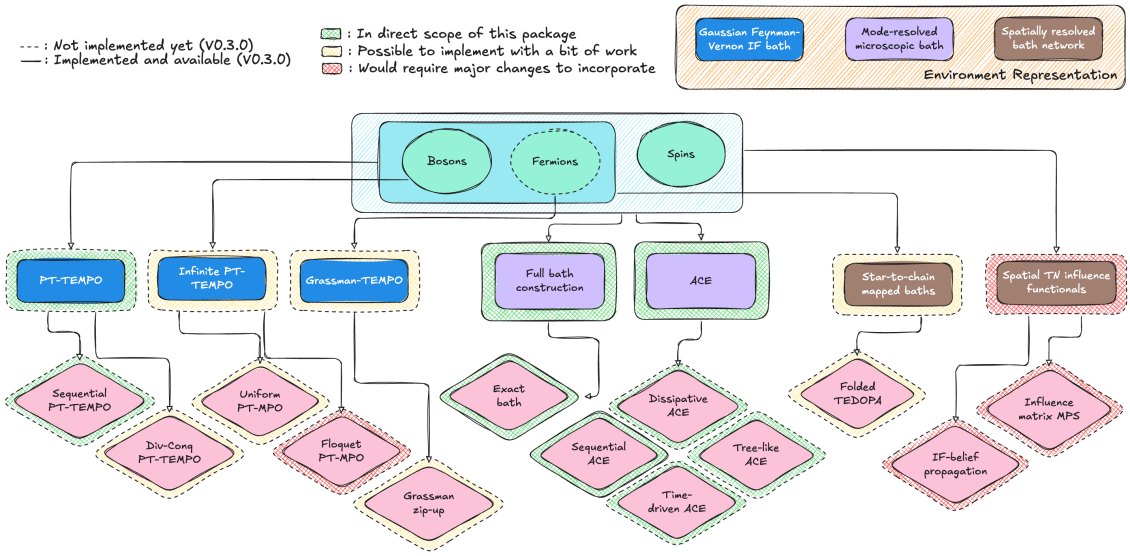


Figure 1: Overview of existing tensor-network routes to construct a process tensor. The diamonds identify construction and evaluation schemes or specialisations within these families; their roles are described in the text. Solid and dashed outlines distinguish whether a method is available in `ProcessTensors.jl` v0.3.0. Green, yellow, and red hatching indicate directions within the package's direct scope, extensions requiring additional development, and approaches requiring major architectural changes, respectively. The branches show representative routes.

Gaussian influence functionals: The first route removes the bath before compressing its memory. For a Gaussian environment with the appropriate linear coupling, the Feynman–Vernon influence functional expresses its effect through two-point bath correlations [10]. The remaining problem is a sum over system histories, with interactions between different times. TEMPO represents this temporal structure using tensor networks [11]; PT-TEMPO reorganises it into a reusable process tensor with the system-operation slots left open [9]. Sequential, divide-and-conquer, and time-translationally invariant finite-time constructions provide different ways of assembling and compressing this network, exploiting repeated temporal structure to reduce the construction effort [12, 13]. For stationary bath correlations, uniTEMPO instead exploits the bath's time translational symmetry and constructs a uniform temporal tensor and boundary vectors, allowing both finite-time evolution and direct access to long-time behaviour through an effective transfer operator [14]. Its direct impact is a much more compressed representation of the process tensor, and its transfer operator can readily be used to obtain steady states. When we wish to drive the system periodically, we can enforce the same time translation symmetry with a modulo of the time period of driving, and obtain Floquet dynamics using the same machinery [15]. Fermionic Gaussian reservoirs require their anticommutation structure to be retained: Grassmann TEMPO supplies a corresponding influence-functional construction, while its zip-up scheme evaluates observables without first forming one large combined temporal tensor [16]. Infinite Grassmann constructions additionally give direct access to steady-state impurity correlations and transport without explicitly following the full transient [17]. Related fermionic influence-functional MPS constructions also address nonequilibrium impurity models [18, 19]. These approaches are particularly useful for structured thermal reservoirs and impurity problems, since the bath can be specified through correlations or spectral data without explicitly retaining every mode. Their efficiency nevertheless depends on temporal compress-

ibility; long memory, strong coupling, and additional coupling channels can make the bonds expensive. Moreover, the Gaussian reduction is a physical restriction, whereas stationarity is an additional restriction of particular acceleration schemes rather than of finite-time influence functionals themselves.

Mode-resolved microscopic environments: The second route starts with the bath degrees of freedom themselves. For a small environment, we can construct the joint propagators directly and trace out the bath while retaining the system's temporal input and output legs ("Exact bath" diamond in Fig. 1). This provides a useful reference construction, although its uncompressed memory space grows with the full bath Liouville space. Automated compression of environments (ACE) makes a more scalable construction possible when the environment consists of independent modes, initially in a product state [20]. Each mode contributes a small process tensor, and their combined influence is compressed as the modes are incorporated. The modes need not be harmonic or Gaussian: spins, truncated oscillators, and more general finite-dimensional constituents can be treated through their microscopic propagators. Sequential ACE adds these contributions one at a time; tree-like ACE merges them hierarchically and uses additional compression sweeps to improve the construction [21]. Time-dependent mode propagators and local dissipative generators extend the same microscopic viewpoint to driven or damped modes [20, 22]. The corresponding diamonds in the figure describe extensions of the local dynamics as well as choices of contraction order. This flexibility is attractive when individual environmental constituents matter or Gaussian assumptions fail. It does not make a large continuum bath inexpensive by default: the mode discretisation, local Hilbert-space cutoffs, time steps, and compression must all be converged. Intermode interactions and initially correlated modes also lie beyond the standard independent-mode construction unless they are incorporated into larger jointly treated units. For a Gaussian continuum, using its analytically known influence functional can therefore avoid work that the more general microscopic route must perform explicitly.

Spatially resolved bath networks: The third route retains how environmental degrees of freedom connect to one another. For linearly coupled free bosonic or fermionic reservoirs, star-to-chain mappings transform the bath into a nearest-neighbour chain; orthogonal-polynomial mappings and TEDOPA established this route to explicit-environment tensor-network simulations [23, 24]. Mapping and evolving the chain alone does not produce a process tensor. To obtain one, the forward and backward evolution networks can be folded together and contracted over the bath while the intervention legs remain open, leaving a tensor network along time [25, 26]. This is the construction route labelled "folded TEDOPA" in the figure. Alternatively, the environment may already be an interacting lattice, with a local subsystem playing the role of the open system. Influence-matrix MPS methods contract the surrounding space-time network in the spatial direction, or solve a spatial self-consistency problem, to encode its effect on the subsystem's histories [25, 27]. Influence-functional belief propagation extends this viewpoint to branching networks by passing temporal influence tensors between neighbouring sites [28]. Boundary influence tensors become a system process tensor only after the appropriate local tensors are attached and the intervention slots retained. These methods give access to interacting, non-Gaussian environments and connect open-system questions to many-body dynamics. Their practical reach is limited by spatial and temporal entanglement, bath-size and local-basis truncations where applicable, and, for loopy networks, the quality of the spatial approximation.

1.2 Software and the scope of this package

Several open-source packages already turn these constructions into useful numerical tools. `OQuPy` implements PT-TEMPO for Gaussian bosonic environments and uses the resulting process tensors for dynamics, multi-time correlations, and control, including extensions to cou-

pled systems [29]. The C++ ACE toolkit provides the independent-mode ACE construction alongside Gaussian-bath PT-MPO algorithms, including divide-and-conquer schemes, with facilities for saving, reusing, and combining environmental tensors [22]. `UniformTEMPO.jl` supplies the uniform construction for infinite process tensors [14, 30], while `GTEMPO.jl` implements Grassmann influence-functional methods for fermionic impurity problems [16, 31, 32]. Within Julia, `QuantumDynamics.jl` offers a broader collection of influence-functional, tensor-network, and other open-system methods [33], and `MPSDynamics.jl` provides explicit-environment simulations based on thermalised TEDOPA, although it does not yet directly use PT-MPO construction [34]. Research codes such as `imcode` provide complementary influence-matrix calculations for lattice and impurity settings [35]. Related path-integral tools, such as `PathSum`, also form part of this wider numerical landscape [36]. For broader comparisons, including methods outside direct PT-MPO construction, we refer readers to the PRX perspective by Keeling, Stoudenmire, Bañuls, and Reichman [37] and the review by Lacroix *et al.* [38].

`ProcessTensors.jl` brings a particular emphasis to this landscape: constructing finite-time process tensors and making their operational content accessible in an intuitive Julia language. We use `ITensors.jl` and `ITensorMPS.jl` as the tensor-network foundation [39]. Named indices and tags let the code follow the same question as a tensor diagram: which legs describe the system, which carry memory, and which should be connected to an intervention? The Liouville-space interface then provides a common language for process tensors, system operations, and their contractions. The present release centres on direct finite-bath construction and sequential ACE, providing both a small-system reference and a compressed route to more general independent-mode environments. With the process available, users can explore preparations, instrument sequences, conditional statistics, and temporal correlations through the same representation. The contribution is this connection between construction, operational use, and an accessible implementation, building on the established algorithms described above.

Figure 1 also places the present implementation within a development roadmap. We describe how alternative constructors can target the same finite-time PT-MPO interface we have, so that improvements in bath construction can be reused by the existing evaluation routines. Nearer extensions include time-dependent and dissipative mode propagators, tree-like ACE, and Gaussian PT-TEMPO construction; divide-and-conquer schemes and chain-based constructions require additional algorithmic machinery. Fermionic support must also account consistently for parity and operator ordering. Floquet process tensors and spatial influence-functional networks require more substantial changes to the representation or construction infrastructure. We provide the details of these extensions in Sec. 7.

The remainder of this paper develops this connection between theory and use step by step. We introduce the Liouville-space and tensor-network conventions in Sec. 2, construct process tensors and their compressed representations in Sec. 3, and in Sec. 4, explain how system operations and instruments extract experimentally meaningful quantities from them. Worked examples in Sec. 5 connect these ingredients to numerical workflows, followed by benchmarks (Sec. 6) and the development roadmap (Sec. 7). Throughout, equations, tensor diagrams, and Julia code are presented together so that the same operation can be followed in each language. The theory and Julia markers in the table of contents indicate the predominant emphasis of each section and help readers choose a route through the material. Students can follow the construction from its physical meaning to a working calculation, while experienced readers can move directly to the implementation codes, examples, and extension points relevant to their own problems.

2 From tensor diagrams to Liouville-space MPS and MPOs

The process tensor is a map acting on other maps. Before constructing such an object, it is useful to place density matrices and quantum operations on equal footing. Moving to the Liouville space does precisely this: density matrices become state-like vectors, while operations on density matrices become ordinary linear operators. This change of language is especially natural for tensor networks, because the same ideas can be read in three equivalent ways—as an equation, as a tensor diagram, and as the corresponding `ITensor` code. We develop these three descriptions together below. Our graphical conventions are adapted from the open-system tensor calculus of Ref. [40], with time and contractions drawn from right to left to match the convention used throughout this paper. In case the readers are not familiar with ITensors and their notations of sites and indices, they are encouraged to read the ITensor documentation [39, 41, 42]. Alternatively, they can also go through the tutorials in the stable documentation of our package.

2.1 A graphical dictionary for tensors

A tensor is represented by a shape with one open wire for every index. In our conventions, the direction of a wire records which dual space that leg belongs to: a leg pointing to the left is a ket index [Fig. 2a], whereas a leg pointing to the right is a bra index [Fig. 2b]. Joining two wires means summing over their shared index. For example, ordinary matrix multiplication,

$$(AB)_{ij} = A_{ik}B_{kj}, \tag{6}$$

is drawn by joining the right-hand wire of A to the left-hand wire of B [Fig. 2e]. Stacking two disconnected tensors vertically, with ket wires on the left and bra wires on the right, represents their tensor product [Fig. 2f]. Closing the two wires of an operator on the same index with a loop gives its trace [Fig. 2g],

$$\mathrm{Tr}(A) = A_{ii}. \tag{7}$$

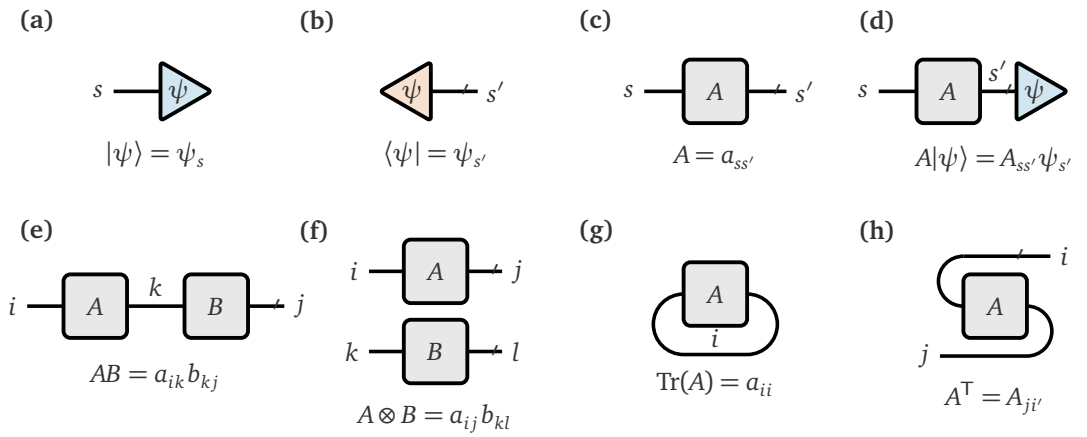


Figure 2: Graphical dictionary used throughout this work. (a) A ket $|\psi\rangle$ with unprimed index s, (b) A bra $\langle\psi|$ with primed index s', (c) An operator A with a ket wire s on the left and a bra wire s' on the right. (d) Contracting the primed wire of A with $|\psi\rangle$. (e) The matrix multiplication $AB = a_{ik}b_{kj}$. (f) The tensor product $A\otimes B = a_{ij}b_{kl}$, drawn by stacking A above B. (g) The trace $\mathrm{Tr}(A) = a_{ii}$. (h) The transpose $A^{\mathsf{T}} = A_{ji}$, obtained by bending the left and right wires to the opposite directions.

Equations (6) and (7), Fig. 2, and Listing 1 express the same basic rule: tensors contract when their indices match. Transposing deserves one warning when we implement it in ITensor

code. For an operator with indices s and s', `swapinds(A, s, prime(s))` exchanges those index identities and therefore implements A^{T}. An ITensor `prime` operation, by contrast, only changes the identity of an index; it does not transpose or conjugate the tensor. Likewise, `dag` conjugates the entries (and reverses index directions when applicable) but does not by itself exchange the two operator indices. This distinction becomes important when we use an unprimed index for the input and a primed copy for its output when constructing operator ITensors.[1]

Julia

```julia
using ITensors

s = Index(2, "s")

psi = ITensor([1.0, 0.0], s) # (a) ket |psi> = psi_s

psidag = dag(prime(psi)) # (b) bra <psi| = psi_s'

A = random_itensor(s, prime(s)) # (c) operator A = a_ss'

Apsi = A * prime(psi) # (d) A|psi> = A_ss' psi_s'

i = Index(2, "i")
k = Index(2, "k")
j = Index(2, "j")
AB = random_itensor(i, k) * random_itensor(k, j) # (e) AB = a_ik b_kj

l = Index(2, "l")
AxB = random_itensor(i, j) * random_itensor(k, l) # (f) A ox B = a_ij
 ↪ b_kl  (no shared index)

TrA = A * delta(s, prime(s)) # (g) Tr(A) = a_ii

AT = swapinds(A, s, prime(s)) # (h) AT = a_ji
```

Listing 1: ITensor codes to realise tensor operations in Fig. 2(a)–(h).

2.2 From operator space to Liouville space

Let $\mathcal{H}$ be a d-dimensional Hilbert space of state vectors. Associated with it is the operator space

$$\mathcal{B}(\mathcal{H}) = \{A : \mathcal{H} \to \mathcal{H}\}, \tag{8}$$

containing all linear operators acting on $\mathcal{H}$. Density matrices, observables, Hamiltonians, and jump operators are therefore different inhabitants of the same space; the construction below is not restricted to Hermitian operators or physical states. The operator space becomes a d^2-dimensional Hilbert space when equipped with the Hilbert–Schmidt inner product

$$\langle\langle B|A\rangle\rangle = \mathrm{Tr}(B^\dagger A). \tag{9}$$

We refer to this operator Hilbert space as the *Liouville space*, $\mathcal{L}(\mathcal{H})$. The name reflects the historical description of dynamics directly in operator space, most notably through the quantum Liouville–von Neumann equation.

[1] Index priming and the operations `dag` and `swapinds` are documented at https://docs.itensor.org/ITensors/stable/IndexType.html and https://docs.itensor.org/ITensors/stable/ITensorType.html.

To use ordinary vector and tensor-network operations in this space, we choose a vector representation

$$A \in \mathcal{B}(\mathcal{H}) \longmapsto |A\rangle\!\rangle \in \mathcal{L}(\mathcal{H}), \tag{10}$$

called *vectorisation*. This does not turn A into a new physical state; it represents the same operator as a vector in Liouville space by rearranging the elements into a vector. In particular, density operators, observables, and non-Hermitian operators can all be vectorised in precisely the same way.

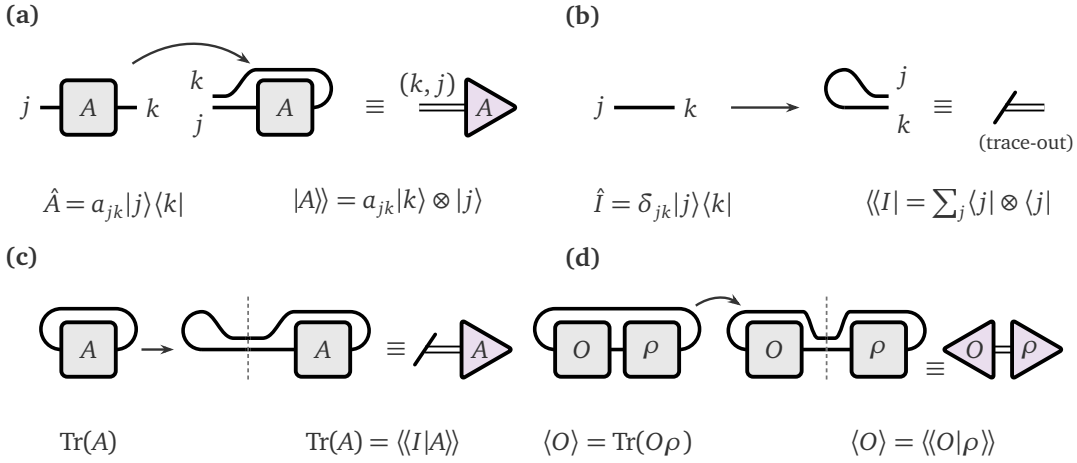


Figure 3: Vectorisation and Liouville-space closures. (a) An operator $\hat{A} = a_{jk}|j\rangle\langle k|$ is vectorised by bending the bra index k over the top, giving column-major order. The fused pair is a double Liouville wire. (b) The identity $\hat{I} = \delta_{jk}|j\rangle\langle k|$ is a straight wire. Bending the left index to the right yields the trace-out operation. (c) The Hilbert-space trace is the Liouville inner product $\mathrm{Tr}(A) = \langle\!\langle I|A\rangle\!\rangle$. (d) An expectation value is the same contraction with an observable, $\langle O\rangle = \langle\!\langle O|\rho\rangle\!\rangle$.

Different vectorisation conventions arrange the matrix entries differently. The package follows Julia's column-major ordering. For $a_{jk} = \langle j|A|k\rangle$, we define

$$|A\rangle\!\rangle \equiv \mathrm{vec}(A) = \sum_{j,k} a_{jk}|k\rangle \otimes |j\rangle, \tag{11}$$

where the first factor records the bra index k (slowest varying index) and the second the ket index j (fastest varying index), and the new double-wired index is a Liouville wire of the tuple (k, j). This is represented as a bent wire with two indices pointing in the same direction in Fig. 3a (note that if we bend the ket index k downwards, that would correspond to a row-major order because the tuple has j as the slowest varying index, and k as the fastest varying index, i.e. the order (j, k)). Equivalently, if we look at the matrix elements of A as a 2D array, the columns of A are stacked vertically in a column-major order:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \longmapsto \begin{pmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{pmatrix}. \tag{12}$$

Thus,

$$\mathrm{vec}\big(|j\rangle\langle k|\big) = |k\rangle \otimes |j\rangle. \tag{13}$$

An important Liouville vector used repeatedly below is the vectorised identity. Graphically, it is represented as a bent wire with two indices pointing in the same direction,

$$|I\rangle\rangle = \sum_j |j\rangle \otimes |j\rangle, \tag{14}$$

as also shown in panel (b) of Fig. 3. Using Eq. (9) and substituting B with the identity operator, we see in Fig. 3(c) that the corresponding trace operation in Liouville space is expressed as an inner product:

$$\mathrm{Tr}(A) = \langle\langle I|A\rangle\rangle. \tag{15}$$

The identity operator, I, is the maximally mixed state up to a normalisation $\rho_{\mathrm{mm}} = I/d$. Vectorising it (panel (b)) graphically yields the operation of closing a leg, or tracing out that subsystem, as illustrated in Fig. 3(b) and (c). More generally, an observable $O = O^\dagger$ closes the same Liouville leg and returns its expectation value, as depicted in Fig. 3(d):

$$\mathrm{Tr}(\rho) = \langle\langle I|\rho\rangle\rangle, \qquad \langle O\rangle = \mathrm{Tr}(O\rho) = \langle\langle O|\rho\rangle\rangle. \tag{16}$$

Leaving the leg unclosed, as in panel (a), instead retains the state-like object $|\rho\rangle\rangle$. These closure rules, visualised in the respective panels of Fig. 3, will reappear when process tensor outputs are traced, measured, or left open.

Listing 2 realises the four panels of Fig. 3 with the package converters: an operator becomes a Liouville vector, the identity becomes the trace-out effect, and traces or expectation values are ordinary inner products.[2]

Julia

```julia
using ProcessTensors

sites = siteinds("S=1/2", 1)
sites_L = liouv_sites(sites)

A = OpSum()
A += 1.0, "Sx", 1
A_L = to_liouville(MPO(A, sites); sites=sites_L) # (a) |A>>

Id = OpSum()
Id += 1.0, "Id", 1
Id_L = to_liouville(MPO(Id, sites); sites=sites_L) # (b) <<I|

TrA = inner(Id_L, A_L) # (c) Tr(A) = <<I|A>>

rho_L = to_liouville(to_dm(MPS(sites, ["Up"])); sites=sites_L)
O = OpSum()
O += 1.0, "Sz", 1
O_L = to_liouville(MPO(O, sites); sites=sites_L)
expect_O = inner(O_L, rho_L) # (d) <O> = <<O|rho>>
```

Listing 2: Vectorisation, the identity, and Liouville closures matching Fig. 3(a)–(d).

[2] `siteinds`, local operators, and `OpSum` are documented at https://docs.itensor.org/ITensorMPS/stable/SiteType.html and https://docs.itensor.org/ITensorMPS/stable/OpSum.html.

2.3 Left and right actions

Vectorisation becomes especially useful when we want to describe an operation such as $A\rho B$ as a linear action on the Liouville-space vector $|\rho\rangle\!\rangle$, because it brings the opeartions on the bra and ket bases to the same space. This allows Hamiltonian evolution, dissipative jumps, and process tensor contractions to be written as ordinary tensor-network operations or MPOs. The only small surprise is that multiplication from the right does not look exactly like multiplication from the left after vectorisation. The transpose appearing below is the bookkeeping rule that keeps the two descriptions consistent.

This viewpoint is closely related to the Choi–Jamiołkowski isomorphism [43,44]. A linear map $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ can be represented in several equivalent ways. It may be described by its action on operators,

$$\rho \longmapsto \Phi(\rho),$$

by a Liouville-space superoperator S_Φ,

$$|\Phi(\rho)\rangle\!\rangle = \mathsf{S}_\Phi |\rho\rangle\!\rangle,$$

or by its Choi operator,

$$J(\Phi) = (\mathcal{I} \otimes \Phi)\big(|I\rangle\!\rangle\langle\!\langle I|\big).$$

The Liouville-space matrix and the Choi operator are two representations of the same linear map, where one can be obtained from the other via index reshuffling. Here we primarily use the Liouville representation, in which maps act on vectorized operators by ordinary matrix multiplication. The Choi representation is useful because it connects properties of maps to properties of operators, as discussed in Sec. 3.2. Introductory material can be found in Refs. [45,46].

The connection between operator multiplication and Liouville-space action is most easily seen by sliding an operator around the same bend used to vectorise the identity in Fig. 3b. For the vectorised identity,

$$(I \otimes A)|I\rangle\!\rangle = (A^{\mathsf{T}} \otimes I)|I\rangle\!\rangle. \tag{17}$$

Graphically, A sits on the ket rail of the cup; after it slides through the bend it reappears transposed on the bra rail [Fig. 4a]. The transpose is not an additional physical operation; it records the change in index ordering induced by our column-major vectorisation convention. Applying this identity to a product gives Roth's lemma,

$$\mathrm{vec}(A\rho B) = (B^{\mathsf{T}} \otimes A)|\rho\rangle\!\rangle. \tag{18}$$

Figure 4b draws the same statement: A, ρ, and B on the ket rail are equivalent to B^{T} and A acting on the two legs of the bent ρ, which after connecting to Fig. 2f, gives the Liouville operator $B^{\mathsf{T}} \otimes A$ on $|\rho\rangle\!\rangle$. Thus the same transformation can be viewed in three equivalent ways,

$$A\rho B \quad \longleftrightarrow \quad \Phi_{A,B}(\rho) = A\rho B \quad \longleftrightarrow \quad \mathsf{S}_{A,B} = B^{\mathsf{T}} \otimes A.$$

The first expression is ordinary operator multiplication, the second is a linear map acting on operator space, and the third is its matrix representation in Liouville space.

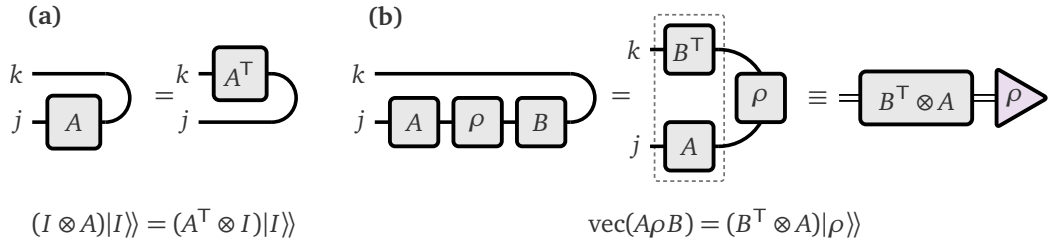


Figure 4: Wire sliding and Roth's lemma. (a) An operator that slides through the cup emerges transposed. (b) The same move on a product yields $\mathrm{vec}(A\rho B) = (B^{\mathsf{T}} \otimes A)|\rho\rangle\!\rangle$.

The two special cases used throughout the package are

$$A\rho \longleftrightarrow (I \otimes A)|\rho\rangle\rangle \equiv A_{\mathrm{L}}|\rho\rangle\rangle, \tag{19}$$

$$\rho B \longleftrightarrow (B^{\mathsf{T}} \otimes I)|\rho\rangle\rangle \equiv B_{\mathrm{R}}|\rho\rangle\rangle. \tag{20}$$

Setting $B = I$ in Eq. (18) leaves A on the ket rail and gives the left action A_{L}, likewise setting $A = I$ leaves B^{T} on the bra rail and gives the right action B_{R}. The suffixes L and R describe multiplication from the left and right of the density operator before vectorisation. Listing 3 realises the two special cases of Fig. 4b, together with the left-right jump that follows from the same identity.

Julia

```julia
using ProcessTensors

sites = siteinds("S=1/2", 1)
sites_L = liouv_sites(sites)
sL = only(sites_L)

A_L = op("Sx_L", sL)         # vec(Sx * rho)
B_R = op("Sz_R", sL)         # vec(rho * Sz)
A_jump = op("S-_Jump", sL)   # vec(S- * rho * S+)
```

Listing 3: Package notation for left, right, and left-right actions.

In this notation, `Sx_L` represents multiplication by S_x from the left, while `Sz_R` represents multiplication by S_z from the right. The `_Jump` suffix represents the corresponding left-right action, also known as the sandwich action in the context of Local Master Equation (LME) superoperators. Together, these local operators provide the elementary building blocks for the Liouville-space dynamics and process tensor constructions developed below.

2.4 Vectorised many-body MPS and Liouvillian MPOs

A single quantum degree of freedom lives in one Hilbert space. A many-body system brings several of them together: if site j has a local Hilbert space $\mathcal{H}_j$ of dimension d_j, then the full state belongs to the joint Hilbert space

$$\mathcal{H} = \bigotimes_{j=1}^{N} \mathcal{H}_j, \qquad \dim \mathcal{H} = \prod_{j=1}^{N} d_j. \tag{21}$$

The price of adding sites is therefore immediate: the number of amplitudes grows exponentially. Tensor networks tame this growth by storing the local degrees of freedom in the form of a matrix-product state (MPS),

$$|\psi\rangle = \sum_{s_1,\ldots,s_N} A_1^{s_1} A_2^{s_2} \cdots A_N^{s_N} |s_1 \cdots s_N\rangle. \tag{22}$$

Neighbouring cores are joined by virtual bonds. Their dimensions control how much correlation and entanglement the MPS can represent across each spatial cut, and truncating them provides the compression that makes the calculation practical. Standard introductions to MPS and matrix-product operators (MPOs) can be found in Refs. [47, 48]; the ITensor implementation used here is described in Ref. [39].[3]

[3] MPS and MPO representations and inner products are documented at https://docs.itensor.org/ITensorMPS/stable/MPSandMPO.html.

A mixed many-body state is described by a density operator on this joint Hilbert space,

$$\rho = \sum_{\boldsymbol{s},\boldsymbol{s}'} \rho_{\boldsymbol{s},\boldsymbol{s}'} |s_1 \cdots s_N\rangle\langle s'_1 \cdots s'_N|, \tag{23}$$

where $\boldsymbol{s} = (s_1, \ldots, s_N)$. When converted to an MPO, each local core now has two physical legs: an unprimed ket index s_j and a primed bra index s'_j, along with the virtual bond indices connecting adjacent cores.

Vectorisation of this density operator can now be performed one site at a time. As shown in Fig. 5, bending each bra wire towards its ket wire produces the column-major pair (s'_j, s_j). A local combiner C_j then fuses this pair into the Liouville index

$$\alpha_j = (s'_j, s_j), \qquad C_j : \mathcal{H}_j^* \otimes \mathcal{H}_j \longrightarrow \mathcal{L}_j, \qquad \dim \mathcal{L}_j = d_j^2. \tag{24}$$

Applying these combiners across the chain turns the `MPO{Hilbert}` representation of ρ into an `MPS{Liouville}` representation of $|\rho\rangle\rangle$. The virtual bonds are untouched; only the two local physical legs are fused. Thus vectorisation changes how the same object is represented, without introducing a second physical state.

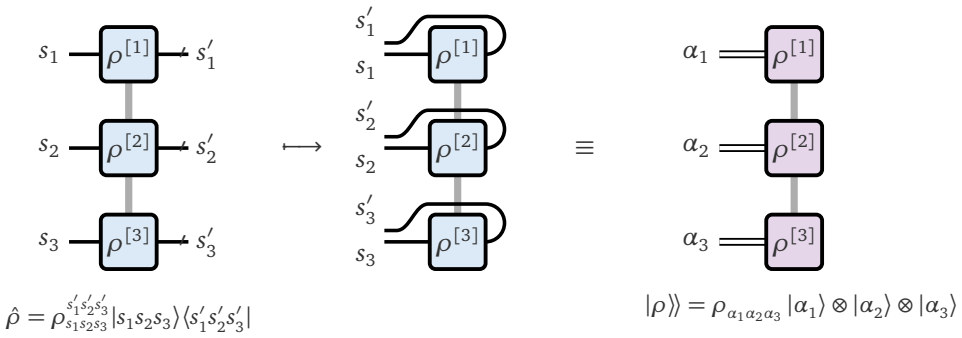


Figure 5: Local many-body vectorisation of a three-site density matrix. A Hilbert-space `MPO{Hilbert}` has an unprimed ket s_j and a primed bra s'_j on every site. Bending each bra over its ket gives the column-major pair (s'_j, s_j), which is fused into the Liouville index α_j of the resulting `MPS{Liouville}`.

In `ProcessTensors.jl`, `to_dm` constructs a Hilbert-space density MPO and `to_liouville` applies the local fusion. The resulting Liouville MPS can then be contracted with vectorised observables or the vectorised identity, exactly as in the single-site discussion.

Julia

```julia
using ProcessTensors

sites = siteinds("S=1/2", 3)
sites_L = liouv_sites(sites)

psi1 = MPS(sites, ["Up", "Up", "Up"])
psi2 = MPS(sites, ["Dn", "Dn", "Dn"])
rho = to_dm([psi1, psi2]; coeffs=[0.8, 0.2])
rho_L = to_liouville(rho; sites=sites_L)

Id = OpSum()
Id += 1.0, "Id", 1
O = OpSum()
```

```
O += 1.0, "Sz", 2

Id_L = to_liouville(MPO(Id, sites); sites=sites_L)
O_L = to_liouville(MPO(O, sites); sites=sites_L)
normalisation = inner(Id_L, rho_L)
expectation = inner(O_L, rho_L)
```

Listing 4: Three-site mixed state matching Fig. 5.

There is a point worth noting while defining the Liouville indices and using them to construct the corresponding Liouville-space object. The Liouville indices should be created once and reused. Calling `liouv_sites` independently produces indices with the same dimensions and tags but different identities (IDs), so ITensor will not contract them as the same legs. The same issue appears when one constructs a density matrix without inputting the `liouv_sites`. This would internally create a new set of indices from the Hilbert-space indices with a different index ID.

For a closed system, the density operator obeys the von Neumann equation

$$\frac{d\rho(t)}{dt} = -i[H,\rho(t)] = -i\big(H\rho(t) - \rho(t)H\big). \tag{25}$$

Using the left and right actions from Eqs. (19) and (20), this becomes one operator acting on a Liouville vector,

$$\frac{d}{dt}|\rho(t)\rangle\!\rangle = \mathcal{L}_H|\rho(t)\rangle\!\rangle, \qquad \mathcal{L}_H = -i\big(H_{\mathrm{L}} - H_{\mathrm{R}}\big). \tag{26}$$

For an `OpSum` $H = \sum_m h_m$, the package constructs this generator term by term: every operator string h_m is converted into its `_L` and `_R` actions before the Liouville MPO is built. It does not first construct the full Hilbert-space Hamiltonian MPO and vectorise that object. This keeps the construction local and preserves the structure already present in the `OpSum`.

Going beyond the unitary time evolution, the most common Markovian open-system generator is the Gorini–Kossakowski–Lindblad–Sudarshan (GKLS) master equation,

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_\mu \gamma_\mu \left(L_\mu \rho L_\mu^\dagger - \frac{1}{2} L_\mu^\dagger L_\mu \rho - \frac{1}{2} \rho L_\mu^\dagger L_\mu \right). \tag{27}$$

The dictionary developed above translates it directly into

$$\mathcal{L} = -i\big(H_{\mathrm{L}} - H_{\mathrm{R}}\big) + \sum_\mu \gamma_\mu \left[(L_\mu)_{\mathrm{Jump}} - \frac{1}{2}(L_\mu^\dagger L_\mu)_{\mathrm{L}} - \frac{1}{2}(L_\mu^\dagger L_\mu)_{\mathrm{R}} \right], \tag{28}$$

where $(L_\mu)_{\mathrm{Jump}} = L_\mu^* \otimes L_\mu$. The package suffixes `_Jump`, `_LdagL_L`, and `_LdagL_R` implement precisely these three dissipative terms.

Julia

```
H = OpSum()
H += 0.7, "Sx", 1
H += 0.2, "Sz", 1

L_H = liouvillian_mpo(H, sites_L)
L_open = liouvillian_mpo(
    H,
    sites_L;
```

```
    jump_ops=[(0.1, "S-", 1)],
)
```

Listing 5: Constructing Hamiltonian and dissipative Liouville MPOs.

We have arrived at a familiar place where we have described the generator of the Liouville-space dynamics in the form of a Liouville MPO. This can be readily used to perform open-quantum system dynamics, analogous to how we use the Hamiltonian MPO to perform unitary time evolution. Over a short interval Δt, the time-evolution operator is given by

$$|\rho(t+\Delta t)\rangle\!\rangle = e^{\Delta t \mathcal{L}} |\rho(t)\rangle\!\rangle. \tag{29}$$

TEBD approximates this propagator by a product of local gates, applied with MPS compression [49–52]. TDVP instead projects the evolution equation onto MPS tangent spaces and integrates the resulting dynamics [52–54]. Both methods implement Liouville-space evolution under a generally non-Hermitian generator. Appendix B gives minimal examples where we can use the existing algorithms in `ITensorMPS.jl`, extend it to the Liouville space, and perform open-quantum dynamics.

Table 1: Hilbert-space expressions, Liouville-space representations, and their `ProcessTensors.jl` counterparts.

Hilbert-space object	Liouville-space object	Package representation
$\|\psi\rangle$	state vector	`MPS{Hilbert}`
ρ	$\|\rho\rangle\!\rangle$	`MPS{Liouville}`
density operator	ket–bra legs fused locally	`to_liouville`
$A\rho$	$A_{\mathrm{L}}\|\rho\rangle\!\rangle$	`"A_L"`
ρA	$A_{\mathrm{R}}\|\rho\rangle\!\rangle$	`"A_R"`
$A\rho B$	$(B^{\mathsf{T}} \otimes A)\|\rho\rangle\!\rangle$	left–right action
$A\rho A^{\dagger}$	$(A^{*} \otimes A)\|\rho\rangle\!\rangle$	`"A_Jump"`
$\mathrm{Tr}\,\rho$	$\langle\!\langle I\|\rho\rangle\!\rangle$	`inner(Id_L,rho_L)`
$\langle O\rangle$	$\langle\!\langle O\|\rho\rangle\!\rangle$	`inner(O_L,rho_L)`
$-i[H,\rho]$	$\mathcal{L}_H\|\rho\rangle\!\rangle$	`liouvillian_mpo`

3 Process tensor theory and construction

The previous section placed density operators, quantum operations, and many-body generators within a common Liouville-space tensor-network language. We now use that language to describe dynamics in which the system may be intervened upon at several times. In this section, we first show why a state-to-state quantum channel is insufficient for such experiments and introduce the process tensor as its multi-time generalisation. We then connect its operational and Choi representations to the temporal MPO stored by `ProcessTensors.jl`, identify the system and memory indices carried by its cores, and assemble the microscopic ingredients required for its construction. Finally, we construct the same `ProcessTensor` representation using the dense and ACE builders, with some details on the compression methods used by `ACE()`, and obtain the Choi MPO representation of the constructed process.

3.1 From quantum channels to multi-time processes

Let us zoom out from the tensor-network details of Sec. 2 for a moment. There, density operators were vectorised, generators were written as many-body MPOs, and their action became an ordinary tensor contraction. Beneath this machinery lies a simple physical requirement: a state evolving from one time to another must remain a valid quantum state. In particular, it must remain Hermitian and positive so that measurement probabilities are real and non-negative, and it must retain unit trace so that those probabilities add up to one. A quantum channel is the linear map that enforces these requirements. Writing

$$\rho' = \Phi[\rho], \qquad \Phi : \mathcal{B}(\mathcal{H}_{\text{in}}) \longrightarrow \mathcal{B}(\mathcal{H}_{\text{out}}), \tag{30}$$

the map Φ is required to be completely positive and trace preserving (CPTP),

$$(\Phi \otimes \mathcal{I}_A)[X] \geq 0 \qquad \text{for every } X \geq 0, \tag{31}$$

$$\text{Tr}(\Phi[\rho]) = \text{Tr}(\rho). \tag{32}$$

The word *complete* is important. Positivity must be preserved not only for an isolated system, but also when the map acts on one part of a larger state entangled with an otherwise untouched ancilla A. Thus, behind the formal acronym "CPTP" are the familiar probability rules of quantum mechanics, together with the demand that they remain consistent when the system is embedded in a larger experiment [8].

We have already encountered one such example, although we did not call it a channel at the time. In Sec. 2, Markovian open-system dynamics was generated by the Gorini–Kossakowski–Lindblad–Sudarshan (GKLS) equation

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \mathcal{L}_t[\rho] = -\mathrm{i}[H(t), \rho] + \sum_\mu \gamma_\mu(t)\left(L_\mu \rho L_\mu^\dagger - \frac{1}{2}\left\{L_\mu^\dagger L_\mu, \rho\right\}\right), \qquad \gamma_\mu(t) \geq 0. \tag{33}$$

Strictly speaking, the Liouvillian $\mathcal{L}_t$ is the generator rather than the channel itself. The corresponding propagator

$$\Phi_{t:s} = \overleftarrow{\mathcal{T}} \exp\left[\int_s^t \mathrm{d}\tau\, \mathcal{L}_\tau\right], \qquad \rho(t) = \Phi_{t:s}[\rho(s)], \tag{34}$$

is CPTP; for a time-independent generator this reduces to $\Phi_{t:s} = \exp[(t-s)\mathcal{L}]$. The Liouvillian MPO introduced previously is therefore the tensor-network representation of the generator whose propagation carries one density operator to another without leaving the space of physical states.

A second, and more general, route to a quantum channel is to regard the apparent non-unitarity of the system as part of a larger unitary evolution. It is straightforward to see that a unitary transformation on a many-body denstiy operator is a CPTP map: they preserve the state norm by definition, $U\rho U^\dagger$ is Hermitian if ρ is Hermitian, and it keeps the same eigenvalues under the transformation, ensuring positivity. If the system begins uncorrelated with an environment in a fixed state ρ_E, then

$$\Phi_{t:s}[\rho_S] = \text{Tr}_E\left[U_{t:s}(\rho_S \otimes \rho_E)U_{t:s}^\dagger\right] = \sum_\mu K_\mu \rho_S K_\mu^\dagger, \tag{35}$$

where the Kraus operators obey

$$\sum_\mu K_\mu^\dagger K_\mu = I_S. \tag{36}$$

In finite dimensions, a linear map is CPTP if and only if it admits such a Kraus representation; equivalently, every quantum channel can be realised by adjoining an initially uncorrelated environment, evolving the joint state unitarily, and discarding the environment afterwards [8]. Unitary evolution and GKLS propagation are therefore not separate ideas, but familiar members of the same family of physical state-to-state maps.

All these channels answer essentially the same question: given the state of the system now, what will its state be later? For a memoryless evolution, this state-to-state picture may be applied successively,

$$\rho_S(t_{k+1}) = \Phi_{k+1:k}[\rho_S(t_k)]. \tag{37}$$

But must the map $\Phi_{k+1:k}$ be determined by the present reduced state alone? Suppose two experimental histories produce exactly the same $\rho_S(t_k)$, while leaving behind different correlations between the system and its environment. An operation performed at an earlier time may have disturbed the environment, conditioned it on a measurement outcome, or stored information in degrees of freedom that are no longer visible in $\rho_S(t_k)$. The same current system state can then have different futures. In that situation, the question "what state comes next?" has omitted part of the experimental input: "how did we reach the present state?".

The natural remedy is to promote the history of interventions to an input of the dynamical description. Let $\mathcal{A}_k$ denote the operation applied to the system at time t_k, while the joint system–environment dynamics between consecutive times is described by $\mathcal{U}_{k+1:k}$. Starting from a possibly correlated state $\rho_{SE}(t_0)$, the final reduced state is

$$\begin{aligned}\rho_S(t_n) &= \mathrm{Tr}_E\Big[\mathcal{U}_{n:n-1}(\mathcal{A}_{n-1}\otimes\mathcal{I}_E)\cdots\mathcal{U}_{1:0}(\mathcal{A}_0\otimes\mathcal{I}_E)[\rho_{SE}(t_0)]\Big] \\ &=: \mathcal{T}_{n:0}\big[\mathcal{A}_{n-1},\ldots,\mathcal{A}_0\big].\end{aligned} \tag{38}$$

The process tensor $\mathcal{T}_{n:0}$ is the higher-order map that remains after the initial joint state, the inaccessible environment, and all intermediate system–environment propagators have been absorbed into a single object [6–8]. A channel accepts a state and returns a state; a one-slot superchannel accepts an operation and returns a state or another operation. A process tensor (also referred to as quantum combs, commonly in the literature) is their multi-time extension: it accepts an ordered sequence of operations and returns the final state. If the final output is also measured or traced, the same object instead returns the associated probability or expectation value.

Despite acting on operations rather than directly on states, a physical process tensor inherits three familiar structural properties from quantum mechanics [6–8]. First, it is *linear in every intervention*. If an operation at time t_k is chosen as a probabilistic mixture $p\mathcal{A}_k + (1-p)\mathcal{B}_k$, then the resulting output is the same mixture of the outputs obtained from $\mathcal{A}_k$ and $\mathcal{B}_k$. For example,

$$\begin{aligned}&\mathcal{T}_{n:0}\big[\ldots, p\mathcal{A}_k + (1-p)\mathcal{B}_k, \ldots\big] \\ &\quad = p\,\mathcal{T}_{n:0}\big[\ldots,\mathcal{A}_k,\ldots\big] + (1-p)\,\mathcal{T}_{n:0}\big[\ldots,\mathcal{B}_k,\ldots\big].\end{aligned} \tag{39}$$

Second, the process tensor is *completely positive*. Inserting completely positive operations at its open times must produce a positive final state, even when those operations act jointly on the system and an arbitrary untouched ancilla. This is the multi-time counterpart of Eq. (31). It guarantees that the process remains physically meaningful when it is embedded inside a larger circuit, rather than only for the isolated sequence used to define it.

Finally, the process tensor is *causal*. A choice made at a later time cannot alter statistics already recorded at an earlier time. Operationally, once the future slots are filled with deterministic trace-preserving operations and their outputs are discarded, the reduced state (or outcome statistics) at time t_k cannot depend on which future operations were chosen:

$$\mathrm{Tr}_{>k}\ \mathcal{T}_{n:0}\big[\mathcal{A}_{n-1},\ldots,\mathcal{A}_0\big] = \mathcal{T}_{k:0}\big[\mathcal{A}_{k-1},\ldots,\mathcal{A}_0\big], \tag{40}$$

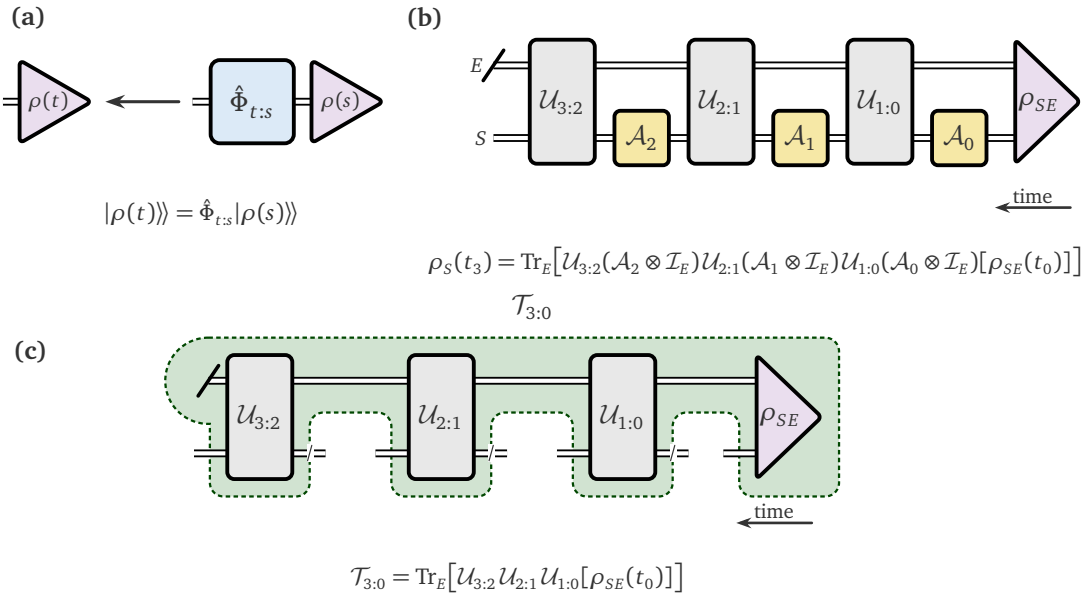


Figure 6: (a) A CPTP map $\hat{\Phi}_{t:s}$ takes a Liouville state $|\rho(s)\rangle\!\rangle$ to $|\rho(t)\rangle\!\rangle$. (b) Joint system–environment evolution with interventions $\mathcal{A}_k$ on the system only; both rails are Liouville wires, the environment is traced out, and the final system leg is left open. Time runs from right to left. (c) The same network with the interventions removed: the remaining object is the process tensor $\mathcal{T}_{3:0}$, with open Liouville slots at each intervention time.

for every k and every pair of future CPTP fillings of the remaining slots. This no-signalling-from-the-future condition also supplies the appropriate multi-time notion of normalisation: every deterministic sequence must preserve total probability. In the Choi representation it becomes a hierarchy of partial-trace constraints, which we introduce in the following subsection. Together, linearity, complete positivity, and causality distinguish a physical time-ordered process from an arbitrary multilinear tensor with matching dimensions.

A point worth emphasising is that an individual trajectory selected from a process tensor need not be deterministic. An operation inserted at an intermediate time may represent a probabilistic event, which would then, reduce the trace of the outcome quantum state. Unit trace is fixed by the instruments contracted with the process, so a process tensor itself is not required to describe a trace-preserving trajectory. We return to this distinction in Sec. 4, where instruments are defined as families of completely positive maps that need not preserve the trace individually, while the complete experiment remains CPTP.

3.2 Choi representation, instruments, and process contraction

The Choi representation recalled in Sec. 2.3 extends naturally to multi-time processes. For the construction used below, we consider an initially uncorrelated system and environment and supply the system preparation $\rho_S(t_0)$ separately. Each dynamical step from t_{k-1} to t_k, with $k = 1, \ldots, n$, has an input space $\mathcal{H}_{\mathrm{in},k}$ and an output space $\mathcal{H}_{\mathrm{out},k}$. The process Choi operator therefore belongs to

$$\Upsilon_{n:0} \in \mathcal{B}\left[\bigotimes_{k=1}^{n}(\mathcal{H}_{\mathrm{in},k} \otimes \mathcal{H}_{\mathrm{out},k})\right].$$

The intervention $\mathcal{A}_k$ connects the output of step k to the input of step $k+1$, so its Choi operator $J(\mathcal{A}_k)$ acts on $\mathcal{H}_{\mathrm{out},k} \otimes \mathcal{H}_{\mathrm{in},k+1}$.

The action of the process tensor in Eq. 38 in terms of its Choi operator is expressed as

$$\rho_S(t_n) = \mathrm{Tr}_{\overline{\mathrm{out},n}}[\Upsilon_{n:0}(\rho_S(t_0)^T \otimes J(\mathcal{A}_1)^T \otimes \cdots \otimes J(\mathcal{A}_{n-1})^T \otimes I)]. \tag{41}$$

Here $\mathrm{Tr}_{\overline{\mathrm{out},n}}$ traces over all subsystems except the final output space $\mathcal{H}_{\mathrm{out},n}$, on which I acts.

The properties discussed in Sec 3.1 translate into conditions on $\Upsilon_{n:0}$. Complete positivity is equivalent to positive semidefiniteness, and the unnormalized Choi convention gives

$$\Upsilon_{n:0} \succeq 0, \qquad \mathrm{Tr}\,\Upsilon_{n:0} = d_S^n, \tag{42}$$

for deterministic dynamics. Causality additionally requires the recursive partial trace conditions

$$\begin{aligned} &\mathrm{Tr}_{\mathrm{out},k}\,\Upsilon_{k:0} = \Upsilon_{k-1:0} \otimes I_{\mathrm{in},k}, \qquad k = 2, \ldots, n, \\ &\mathrm{Tr}_{\mathrm{out},1}\,\Upsilon_{1:0} = I_{\mathrm{in},1}. \end{aligned} \tag{43}$$

These constraints impose that later interventions cannot affect past statistics and ensure normalization under trace-preserving interventions.

Instruments, outcome-resolved probabilities, and the numerical contraction of a schedule against a stored process tensor are developed in Sec. 4. The object stored by the package is not the dense Choi matrix, but its Liouville-space matrix-product representation, which we turn to now.

3.3 Anatomy of a `ProcessTensor`

The Choi representation introduced above is complete, but its size grows exponentially with the number of intervention times. This is the temporal counterpart of the many-body problem encountered in Sec. 2: writing down every possible history as one dense tensor quickly becomes impractical. In this package, the process tensor is represented in matrix-product operator form. The environmental degrees of freedom are compressed into bond indices that carry only the influence of the environment that remains relevant to the system at each timestep. The outcome is an MPO whose physical indices are ordered in time rather than in space; we write this stored object as $\Upsilon_{n:0}$, shown in Fig. 7,

$$\left[\Upsilon_{n:0}\right]_{\boldsymbol{o},\boldsymbol{i}} = \sum_{\chi_1,\ldots,\chi_{n-1}} [Q^{[n]}]^{\chi_{n-1}}_{o_n i_n} [Q^{[n-1]}]^{\chi_{n-1}\chi_{n-2}}_{o_{n-1} i_{n-1}} \cdots [Q^{[k]}]^{\chi_k \chi_{k-1}}_{o_k i_k} \cdots [Q^{[1]}]^{\chi_1}_{o_1 i_1} \tag{44}$$

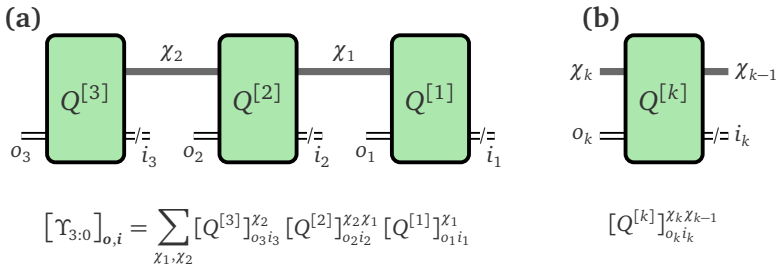


Figure 7: Matrix-product representation of a process tensor. (a) A three-step PT-MPO with time running from right to left: each core $Q^{[k]}$ carries a pair of double-wired system legs (output o_k, input i_k) and neighbouring cores are joined by a thick memory link χ_k. (b) Index anatomy of a generic interior core. Open system legs mark the times at which operations may be inserted, while the horizontal bonds transmit the compressed influence of the environment between different parts of the experiment.

In the present implementation, core $Q^{[k]}$, $k = 1, \ldots, n$, propagates the system over $[t_{k-1}, t_k]$ and carries input and output legs i_k, o_k, both tagged `tstep=k`; instruments between consecutive cores connect o_k to i_{k+1}. Here i_k and o_k are the system input and output indices of that core, while χ_k is an internal memory index joining neighbouring cores. The bond dimensions $D_k = \dim(\chi_k)$ quantify how large an effective space must be carried across each temporal cut. A Markovian process factorises into independent channels, so its Choi operator is a tensor product of ordinary single-step Choi operators and the corresponding PT-MPO needs only trivial memory links. A non-Markovian process generally has $D_k > 1$. A small value of D_k does not imply the absence of memory; it means that the relevant memory is compressible [6, 9].

To illustrate the indices that exist in our Process Tensor MPO (PT-MPO), let us consider a three-step dense process tensor with one bath spin and a system spin (we will see how to construct this in the next subsection). The first core contains the following site indices:

Julia

```julia
siteinds(pt, 1)
```

Terminal

```
2-element Vector{Index{Int64}}:
 (dim=4|id=810|"Liouv,n=1,ptype=S=1/2,tstep=1")'
 (dim=4|id=810|"Liouv,n=1,ptype=S=1/2,tstep=1")
```

Listing 6: Site indices of the first PT-MPO core.

The primed index is i_1, the unprimed index is o_1, and their common `id` shows that they are two prime levels of the same ITensor index. The numerical ID is generated at runtime and can change from what is shown above.

The temporal memory indices are equally direct:

Julia

```julia
linkinds(pt)
```

Terminal

```
2-element Vector{Index{Int64}}:
 (dim=4|id=388|"Link,PT,tstep=1")
 (dim=4|id=209|"Link,PT,tstep=2")
```

Listing 7: Memory-link indices of the three-step process tensor.

These tags distinguish a PT-MPO from a conventional spatial MPO. A physical PT leg has `Liouv` (Liouville rather than Hilbert space), `ptype=S=1/2` (the underlying physical site type), and `tstep=k` (temporal position); its prime level distinguishes input from output. A memory bond has `PT` and `tstep=k` in addition to `Link`, replacing the usual spatial link position. Here each bond has dimension 4, the retained bath Liouville dimension; under compression the effective dimensions $D_k = \dim(\chi_k)$ may vary with k.

Unknown property access is delegated to `core`, so familiar tensor-network inspection functions can be applied directly to `pt`. For example, `linkdims(pt)` and `maxlinkdim(pt)` inspect the temporal memory bonds. The package-specific helpers `input_sites`, `output_sites`, and `coupling_times` then expose the exact temporal legs on which instruments will act. We postpone those contractions to the later sections.

Finally, connecting this object back to its Choi representation, the Liouville legs i_k and o_k are simply the column-major packing of the Hilbert-space Choi indices of Sec. 3.2. Let a_k (a'_k) label the unprimed (primed) basis of $\mathcal{H}_{\text{in},k}$ and b_k (b'_k) likewise for $\mathcal{H}_{\text{out},k}$. Eq. (11) then identifies

$$i_k = (a'_k, a_k), \qquad o_k = (b'_k, b_k), \tag{45}$$

and the two representations of the same process are related by the reshuffling

$$\left[\Upsilon^{\text{Choi}}_{n:0}\right]_{(a_1 b_1 \ldots a_n b_n),(a'_1 b'_1 \ldots a'_n b'_n)} = \left[\Upsilon^{\text{Liouville}}_{n:0}\right]_{(o_1 \ldots o_n),(i_1 \ldots i_n)}. \tag{46}$$

ITensor primes on a PT core distinguish the input leg i_k from the output leg o_k; they should not be confused with the Hilbert-space primes packed inside each Liouville index.

3.4 Ingredients to construct a `ProcessTensor`

Having identified the anatomy of a `ProcessTensor`, we now specify the microscopic data needed to construct it. We introduce these inputs in the same order in which they define the physics: first the accessible system and its free Hamiltonian, then the environment together with its initial state, free Hamiltonian, and coupling to the system, and finally the discrete time grid on which the process is represented. Throughout, we use one system spin coupled to one bath spin for three time steps—the smallest running example that exhibits a non-trivial memory bond—so that each ingredient can be followed directly into the dense construction in the next subsection.

1. System: The first ingredient is the accessible system. The PT-MPO constructor infers the type of particle that the system represents from the site index used to construct it. In v0.3.0 of `ProcessTensors.jl`, the supported system site types are `Spin-1/2`, `Spin-1`, `Qubit`, and `Boson`. The system's free Hamiltonian is also used by the constructor to append the free evolution of the system to the PT-MPO. In our small example, we use a single `Spin-1/2` system with the Hamiltonian

$$H_S = h S^x_S, \qquad h = 0.7. \tag{47}$$

It is constructed from an ordinary Hilbert-space spin index:

Julia

```julia
system_sites = siteinds("S=1/2", 1)

h = 0.7
H_S = OpSum()
H_S += h, "Sx", 1

system = spin_system(system_sites, H_S)
println(system)
```

Terminal

```
ProcessTensors.SpinSystem
  sites: 1
  space: Liouville
  site dims: 4
  dissipative: false
```

Listing 8: System Hamiltonian and `SpinSystem` constructor.

The first line identifies the model type. Although we supplied a Hilbert index of dimension 2, `spin_system` converted it internally into a Liouville index of dimension $2^2 = 4$; this accounts for the next three lines. The final line reports that no Lindblad jump operators were supplied. The present process tensor builder requires a single-site system because the free-system propagation is embedded into every temporal core. An initial system state is intentionally absent: it is an experimental preparation and will later be inserted into the first open process tensor leg as an instrument.

2. Environment: The next ingredient is the non-Markovian environment and its microscopic content that dictates how it influences the system. We store this microscopic content in an `AbstractBathMode`. An environment is a collection of individual bath modes, and it may be empty. In v0.3.0 of `ProcessTensors.jl`, the supported bath-mode site types are `Spin-1/2`, `Spin-1`, and `Boson`. For each bath mode we need its Liouville site, local Hamiltonian, initial state, and coupling to the system. The present implementation does not include initially correlated bath states. In our example, we take

$$H_E = \frac{\omega_E}{2} S_E^z, \qquad H_{SE} = g S_E^z S_S^z, \qquad \rho_E(0) = |\Downarrow\rangle\langle\Downarrow|, \tag{48}$$

with $\omega_E = 1.1$ and $g = 0.35$:

Julia

```julia
bath_sites = siteinds("S=1/2", 1)
bath_sites_L = liouv_sites(bath_sites)

omega_E = 1.1
H_E = OpSum()
H_E += (omega_E / 2), "Sz", 1

psi_E0 = MPS(bath_sites, ["Dn"])
rho_E0 = to_dm(psi_E0)
rho_E0_L = to_liouville(rho_E0; sites=bath_sites_L)

g = 0.35
H_SE = OpSum()
H_SE += g, "Sz", 1, "Sz", 2

mode = spin_mode(
    bath_sites_L,
    H_E,
    rho_E0_L;
    coupling=H_SE,
)
environment = spin_bath([mode])

println(environment)
```

Terminal

```
ProcessTensors.SpinBath
  modes: 1
  space: Liouville
  site dims: 4
  bath Liouville dimension: 4
```

```
mode summary:
  [1] SpinMode(dim=4, H_terms=1, coupling_terms=1)
```

Listing 9: Bath Hamiltonian, initial state, coupling, and `SpinBath` constructor.

The two site labels in H_{SE} follow a local convention used by every independent bath mode: site 1 is the bath mode and site 2 is the system coupling site. They do not refer to two sites inside the system. This output first identifies a spin bath containing one mode. Its single Liouville site again has dimension 4, so the complete bath Liouville space also has dimension $D_E = 4$. The mode summary confirms that both the local bath Hamiltonian and the mode–system coupling contain one `OpSum` term. Notice that $\rho_E(0)$ was vectorised on the exact indices stored in `bath_sites_L`. Two Liouville indices with the same dimension and tags are not interchangeable if they have different ITensor identities.

3. Time grid: The last physical ingredient is the time grid,

$$t_k = k\Delta t, \qquad k = 0, \ldots, n_{\text{steps}}, \tag{49}$$

which is specified by

Julia

```julia
dt = 0.05
nsteps = 3
t_final = round(dt * nsteps; digits=10)
println("time grid: dt=$dt, nsteps=$nsteps, t_final=$t_final")
```

Terminal

```
time grid: dt=0.05, nsteps=3, t_final=0.15
```

Listing 10: Time grid for the dense process tensor example.

Process labels count the stored intervention slots, not the number of time stamps. An `nsteps`$= N$ process contains the cores $Q^{[1]}, \ldots, Q^{[N]}$, tagged `tstep` $= 1, \ldots, N$, and is denoted $\Upsilon_{N:0}$; `t_final`$= N\,\Delta t$ is the endpoint of the last propagation interval. The present example is therefore `nsteps` $= 3 \iff Q^{[1]}, Q^{[2]}, Q^{[3]} \iff \Upsilon_{3:0}$, with $t_3 =$ `t_final` $= 0.15$. Fig. 8 summarises how these ingredients combine: a single-site system, an environment assembled from bath modes, and the chosen time grid.

3.5 Builders to construct a `ProcessTensor`

The ingredients of Sec. 3.4 can be assembled by two builders that return the same `ProcessTensor` representation. `Dense()` retains the complete environmental Liouville space and suits one mode, a small multimode bath, or an exact reference calculation; `ACE()` adds independent modes sequentially and compresses the temporal memory after each addition.

For the one-spin environment introduced above, the complete joint Liouville space is still sufficiently small for the `Dense()` builder. To see what this method constructs, we return to the operational definition in Eq. (38). Between two consecutive intervention times, the system and environment evolve together under the one-step map

$$\mathcal{G}_{S+E}(\Delta t) = \exp\left[\Delta t\,(\mathcal{L}_S + \mathcal{L}_E + \mathcal{L}_{SE})\right]. \tag{50}$$

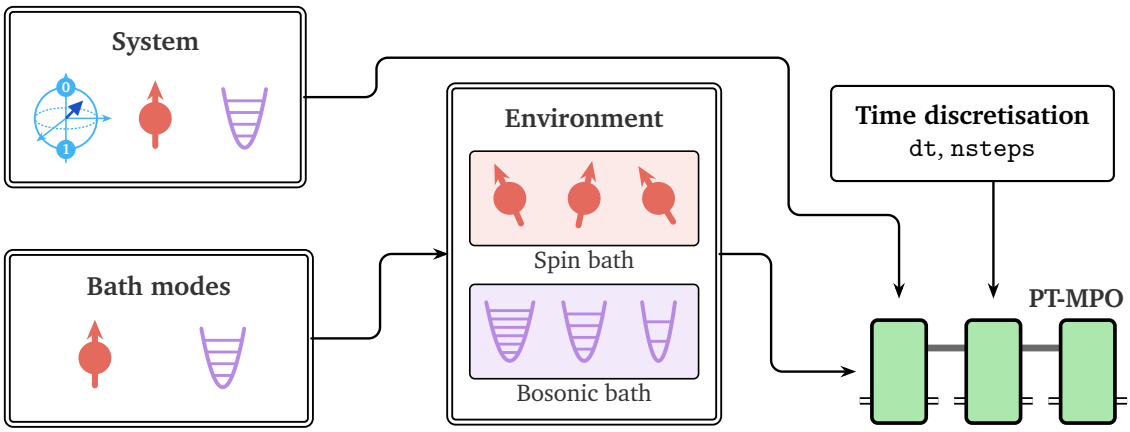


Figure 8: Microscopic ingredients of a `ProcessTensor`.

Constructing the process tensor amounts to repeating this map across the chosen time grid while leaving the system input and output legs at every step open. The initial environmental leg is contracted with the supplied state $|\rho_E(0)\rangle\!\rangle$, the environmental output of one step is connected to the environmental input of the next, and the final environmental leg is contracted with the trace effect $\langle\!\langle I_E|$. These contractions implement precisely the initialization and final partial trace appearing in Eq. (38); the system legs remain open because the corresponding preparations and interventions will be supplied later.

The environmental legs connecting successive propagators are the microscopic origin of the memory links introduced in Eq. (44). In a dense construction, they retain the complete bath Liouville space. A bath with Hilbert-space dimension d_E therefore produces uncompressed memory links of dimension d_E^2. For N_E modes with local dimensions d_m, the complete joint Liouville dimension is

$$D_{S+E}^{(L)} = d_S^2 \prod_{m=1}^{N_E} d_m^2, \tag{51}$$

which makes direct exponentiation practical only for a single mode or a small multimode environment.

Although Eq. (50) gives the complete physical propagator, the package does not construct it as one inseparable exponential. Instead, it writes

$$\mathcal{L}_{S+E} = \mathcal{L}_S + \mathcal{L}_{E+SE}, \qquad \mathcal{L}_{E+SE} := \mathcal{L}_E + \mathcal{L}_{SE}. \tag{52}$$

This separation reflects the architecture shared by the process tensor builders. The bath and interaction generator $\mathcal{L}_{E+SE}$ constructs the environmental influence and its temporal memory links, whereas $\mathcal{L}_S$ generates the deterministic free evolution of the accessible system. Free-system evolution does not itself create environmental memory, but it changes how the system interacts with that memory and must therefore remain part of the final process tensor. Accordingly, `ProcessTensors.jl` constructs the environmental slab first and embeds the free-system map into every PT-MPO core afterwards; the system evolution is separated during construction, not discarded.

Because $\mathcal{L}_S$ and $\mathcal{L}_{E+SE}$ need not commute, their propagators cannot generally be multiplied without approximation. Defining

$$\mathcal{M}_S(\tau) = \exp(\tau\mathcal{L}_S), \qquad \mathcal{Q}_{E+SE}(\tau) = \exp(\tau\mathcal{L}_{E+SE}), \tag{53}$$

the exact one-step map and the two available splittings are

$$\begin{aligned}\mathcal{G}_{S+E}(\Delta t) &= \exp\left[\Delta t\,(\mathcal{L}_S + \mathcal{L}_{E+SE})\right] \\ &\approx \begin{cases} \mathcal{Q}_{E+SE}(\Delta t)\,\mathcal{M}_S(\Delta t) & \text{for } \texttt{Trotter\{1\}()}, \\ \mathcal{M}_S(\Delta t/2)\,\mathcal{Q}_{E+SE}(\Delta t)\,\mathcal{M}_S(\Delta t/2) & \text{for } \texttt{Trotter\{2\}()}. \end{cases}\end{aligned} \tag{54}$$

The first line is the first-order Lie–Trotter splitting; because it applies the two maps in a fixed order, we refer to it as the asymmetric choice. The second is the Strang splitting, the familiar symmetric second-order formula [55, 56]. For Hamiltonian evolution, $\mathcal{M}_S(\tau)$ is the unitary channel generated by $U_S(\tau) = \exp(-\mathrm{i}H_S\tau)$; when the system contains Lindblad terms, it denotes the corresponding free-system CPTP map. The symmetric choice reduces the local splitting error from $\mathcal{O}(\Delta t^2)$ to $\mathcal{O}(\Delta t^3)$, and hence the accumulated error at fixed final time from $\mathcal{O}(\Delta t)$ to $\mathcal{O}(\Delta t^2)$. It does not eliminate time-discretisation error, so physical results will be more accurate as Δt is decreased.

With this construction, the process tensor for the one-spin example is obtained by

Julia

```julia
using ITensors.Ops: Trotter

pt_dense = build_process_tensor(
    system;
    method=Dense(),
    environment=environment,
    dt=dt,
    nsteps=nsteps,
    sys_alg=Trotter{2}(),
    progress=false,
)

println(pt_dense)
```

Terminal

```
3-step ProcessTensor{SpinSystem, SpinBath} | dt=0.05 | t_final=0.15 |
↪  maxlinkdim=4

  system:      SpinSystem(nsites=1, dissipative=false)
  environment: SpinBath(nmodes=1, D_bath=4, coupling=true)
  core:        MPO{Liouville}(length=3, linkdims=[4, 4])
```

Listing 11: Dense construction of the one-spin-bath process tensor.

The three MPO tensors correspond to the three propagation intervals specified by `nsteps`. Each memory link has dimension $4 = 2^2$, confirming that the dense builder has retained the complete Liouville space of the single bath spin. No temporal compression has been introduced at this stage.

The dense product space becomes impractical for many microscopic modes. However, when the environment constructed in Sec. 3.4 is a collection of initially uncorrelated modes, each with its own Hamiltonian and system coupling but no direct mode–mode coupling, its

generator separates as

$$\mathcal{L}_{E+SE} = \sum_{m=1}^{N_E} \mathcal{L}_{E_m+SE_m}, \tag{55}$$

This star-topology environment structure lets `ACE()` propagate each mode in its small Hilbert space, convert it to a closed influence PT-MPO, and absorb it sequentially without ever forming the full dense environmental operator.

After joining mode m, ACE compresses the temporal links by SVD:

$$\widetilde{\Upsilon}^{(m)}_{n:0} = \mathcal{C}_{\varepsilon,\chi_{\max}}\left[\widetilde{\Upsilon}^{(m-1)}_{n:0} \star \Upsilon^{(m)}_{n:0}\right], \tag{56}$$

Here $\star$ joins the accumulated and single-mode influences and ε (the `cutoff`) and $\chi_{\max}$ (the `maxdim`) are the singular value threshold and maximum bond dimension, respectively. When their generators do not commute, `combine_alg=Trotter{2}()` performs this join with symmetric second-order ordering. The map $\mathcal{C}_{\varepsilon,\chi_{\max}}$ then compresses each temporal cut with

$$\sigma_j > \varepsilon\sigma_1, \qquad \chi_k \leq \chi_{\max}. \tag{57}$$

Thus χ_k labels an effective temporal state rather than the complete microscopic bath, and ACE is efficient when its dimension remains small. A more detailed derivation and applications of the ACE algorithm can be found in Refs. [20, 22, 38].

Two compression schedules are available. The default `:canonzip` first joins a complete mode without truncation, canonicalises towards the final time, and performs a right-to-left relative-SVD sweep, allowing the retained basis to see the mode over the full interval. `:zipup_cpp` truncates during the forward join before a backward sweep, reducing intermediate sizes for long chains or tight memory budgets at the cost of greater cutoff sensitivity. Tree-like joining is another ACE strategy [21], but is not implemented here.

For a vector `modes` whose elements were constructed as in the preceding subsection, the many-mode process is built with

Julia

```julia
ace_environment = spin_bath(modes)

pt_ace = build_process_tensor(
    system;
    method=ACE(
        cutoff=1e-10,
        maxdim=512,
        compression=:canonzip,
    ),
    environment=ace_environment,
    dt=dt,
    nsteps=nsteps,
    sys_alg=Trotter{2}(),
    combine_alg=Trotter{2}(),
)
```

Listing 12: ACE construction for a bath of independent microscopic modes.

The cutoff is a convergence parameter, not a target observable accuracy: increasing it discards more directions, while a smaller `maxdim` may force truncation above the cutoff. Because generic SVD truncation need not preserve complete positivity or causal normalisation,

aggressive compression can cause trace drift, small negative eigenvalues, or unstable observables. Convergence therefore requires smaller cutoffs and larger caps while monitoring trace, Hermiticity, minimum eigenvalue, and target observables.

We can obtain the Choi operator defined in Sec. 3.2 from a `ProcessTensor` using the function `choi_pt_mpo`.

Julia

```julia
W = choi_pt_mpo(pt_dense)

j = 1
i = only(siteinds(W, j; tags="in", plev=0))
o = only(siteinds(W, j; tags="out", plev=0))
println((i, prime(i), o, prime(o)))
println(tr(W))
```

Terminal

```
((dim=2|id=95|"Choi,Site,in,tstep=1"),
 ↪  (dim=2|id=95|"Choi,Site,in,tstep=1")',
 ↪  (dim=2|id=556|"Choi,Site,out,tstep=1"),
 ↪  (dim=2|id=556|"Choi,Site,out,tstep=1")')
8.0 + 0.0im
```

Listing 13: Choi MPO of the dense one-spin process tensor.

Here W is the unnormalized Choi operator, returned as an `MPO{Hilbert}` with one core for each propagation interval. On each core there are four Hilbert indices of dimension d_S: two for the input and output legs, and two for the ket and bra. The tags `in` and `out` label the input and output ports, the unprimed index is ket-facing, and the primed index is bra-facing. A point worth noting here is that priming changes its role: on the stored PT-MPO a prime separates the input leg from the output leg, whereas on W a prime separates the bra-facing index from the ket-facing index.

4 Quantum instruments and multi-time experiments

Constructing a process tensor is only the first half of the story; the physics appears from it when we construct and contract the intermediate interventions of the system at each timestep. The process tensor constructed in the previous section is deliberately left as an abstract object with its system input and output legs open for later intervention. This is what makes the object reusable. The same PT-MPO can describe uninterrupted reduced dynamics, a controlled experiment, a post-selected measurement trajectory, or a multi-time correlation function; what changes between these calculations is not the process tensor, but the instruments contracted with its temporal legs. We first introduce these instruments in their strict operational sense, then explain the instrument interface and classification in `ProcessTensors.jl`, the lazy schedules that organise them, and the memory-bearing testers that correlate controls across time. The section concludes with two key functions that turn these ingredients into physical results: `evaluate_process` and `evolve`.

4.1 Operational meaning of instruments

A quantum operation describes one possible transformation of a system, including the possibility that the transformation occurs only for a selected measurement outcome. For an outcome x, it is a completely positive, trace-non-increasing map

$$\mathcal{A}_x : \mathcal{B}(\mathcal{H}_{\text{in}}) \longrightarrow \mathcal{B}(\mathcal{H}_{\text{out}}), \qquad \mathcal{A}_x(\rho) = \sum_\mu K_{x\mu} \rho K_{x\mu}^\dagger, \tag{58}$$

with

$$\sum_\mu K_{x\mu}^\dagger K_{x\mu} \preceq \mathbb{I}. \tag{59}$$

Here, μ labels the unresolved Kraus components associated with that outcome. Complete positivity guarantees that

$$(\mathcal{A}_x \otimes \mathcal{I}_R)(X) \geq 0 \qquad \text{for every } X \geq 0 \tag{60}$$

and for an arbitrarily large untouched reference R. This requirement is essential in a multi-time setting: the system on which an intervention acts may already be correlated with an environment, an ancillary memory, or other degrees of freedom that are not part of the local description. The inequality in Eq. (59) then ensures that the trace of the unnormalised output can be interpreted as the probability of the selected event,

$$p(x) = \text{Tr}[\mathcal{A}_x(\rho)], \qquad \rho_x = \frac{\mathcal{A}_x(\rho)}{p(x)} \quad \text{when } p(x) > 0. \tag{61}$$

A quantum instrument is the complete family of such outcome maps,

$$\mathfrak{I} = \{\mathcal{A}_x\}_{x \in \mathsf{X}}, \qquad \sum_{x \in \mathsf{X}} \mathcal{A}_x \text{ is completely positive and trace preserving,} \tag{62}$$

or, equivalently,

$$\sum_{x,\mu} K_{x\mu}^\dagger K_{x\mu} = \mathbb{I}. \tag{63}$$

The selected map $\mathcal{A}_x$ is generally probabilistic and trace-non-increasing, while the non-selective operation $\sum_x \mathcal{A}_x$, obtained by forgetting the outcome, is deterministic and trace preserving. This distinction between an individual event and the complete experiment is the central operational content of the instrument formalism [57, 58].

Most familiar laboratory operations fit into this hierarchy. An initial preparation has no incoming quantum state and is formally represented as a map from the one-dimensional space $\mathbb{C}$ to the prepared state, $\mathcal{P}_{\rho_0}(\lambda) = \lambda\rho_0$, where λ is a scalar weight, normally equal to one for a deterministic preparation. A unitary pulse and a noise channel are deterministic two-sided maps from one system state to another. A measurement outcome x, by contrast, is represented by a positive operator E_x, called a quantum effect. The effects associated with all possible outcomes form a positive operator-valued measure (POVM),

$$E_x \geq 0, \qquad \sum_x E_x = \mathbb{I}, \tag{64}$$

so each E_x is referred to, henceforth, as a POVM element and satisfies $0 \leq E_x \leq \mathbb{I}$. For an input state ρ, this POVM element assigns the outcome probability

$$p(x) = \text{Tr}(E_x \rho).$$

Combining the POVM element with a newly prepared state σ_x gives the measure-and-prepare operation

$$\mathcal{A}_x^{\mathrm{MP}}(\rho) = \mathrm{Tr}(E_x\rho), \sigma_x, \tag{65}$$

which explicitly breaks the system channel carrying information from the past while allowing the environment to retain a conditional memory of the outcome.

An observable should be distinguished from an instrument outcome. A positive effect E_x defines a physical event, whereas a general Hermitian observable $O = \sum_x o_x E_x$ defines the weighted combination

$$\langle O \rangle = \sum_x o_x p(x). \tag{66}$$

The direct insertion of O into a tensor-network contraction is therefore a convenient linear probe, not necessarily a completely positive outcome map. The package supports both uses: physical operations are used to describe experimental trajectories, while general operator insertions efficiently evaluate expectation values and correlation functions.

For a sequence of outcomes $\boldsymbol{x} = (x_0, \ldots, x_{N-1})$, the process tensor in Eq. (38) returns the unnormalised conditional state

$$\tilde{\rho}_S^{(\boldsymbol{x})}(t_N) = \mathcal{T}_{N:0}\big[\mathcal{A}_{x_{N-1}}^{(N-1)}, \ldots, \mathcal{A}_{x_0}^{(0)}\big], \qquad p(\boldsymbol{x}) = \mathrm{Tr}\,\tilde{\rho}_S^{(\boldsymbol{x})}(t_N). \tag{67}$$

Summing over every outcome at every time replaces the selected maps by deterministic CPTP operations and restores unit total probability. Thus, a process tensor itself is causally normalised, but a trajectory obtained from it need not have unit trace. This is precisely the distinction anticipated at the end of Sec. 3.1 and formalised by the operational process tensor framework [6].

4.2 Instrument organisation in `ProcessTensors.jl`

The package uses the word *instrument* slightly more broadly than the strict definition in Eq. (62). An `AbstractInstrument` is any lazily specified tensor that can be inserted on the system legs of a `ProcessTensor`. This interface includes physical preparations and channels, but also observable insertions, arbitrary left–right actions, open-leg instructions, and user-supplied tensors. The common interface is computational: it tells the contraction how to materialise an object on the correct PT indices. As of v0.3.0, complete positivity and trace preservation are not inferred for an arbitrary custom tensor, so users who construct a physical protocol remain responsible for supplying physical maps.

Two abstract subtypes encode the first structural distinction: `SingleLegInstrument` and `TwoLegInstrument`. The first occupies one process tensor leg and therefore begins or terminates a system wire. The second joins an output of one PT core to the input of the next and acts as a map between adjacent intervention slots. Using the notation of Eq. (44), the unprimed system index o_k is the output of the core $Q^{[k]}$, whereas the primed index i_k is its input. Their ITensor tags contain the same `tstep=k`; their prime levels distinguish the two roles,

$$\mathrm{plev}(o_k) = 0, \qquad \mathrm{plev}(i_k) = 1. \tag{68}$$

A between-attachment at anchor k connects o_k to i_{k+1},

$$[\mathcal{A}_k]_{i_{k+1}o_k}, \tag{69}$$

while the initial preparation and terminal effect have only one leg,

$$[\mathcal{P}_{\rho_0}]_{i_1} = |\rho_0\rangle\!\rangle_{i_1}, \qquad [\mathcal{M}]_{o_N} = \langle\!\langle M|_{o_N}. \tag{70}$$

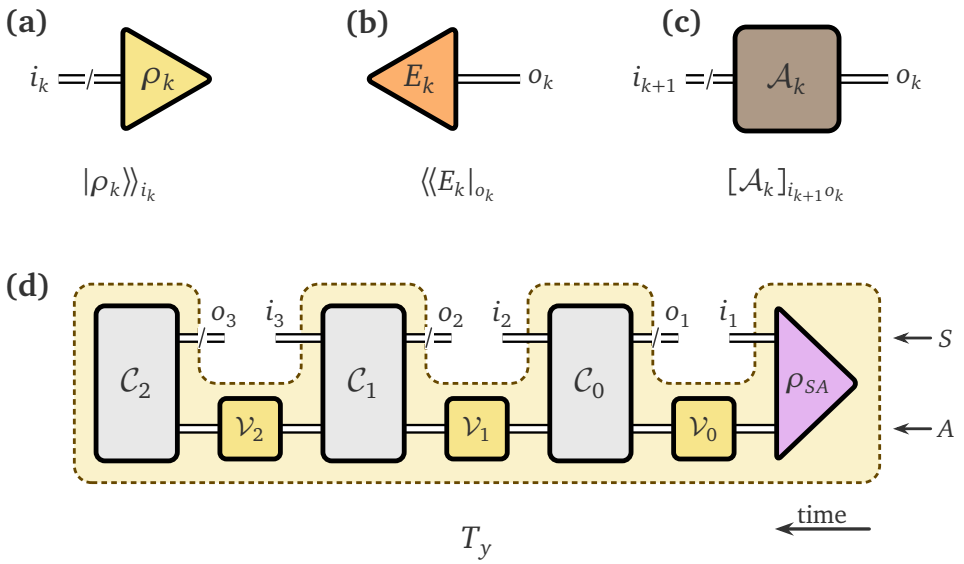


Figure 9: Anatomy of the instrument objects contracted with a PT-MPO. (a) An input-only preparation occupies the primed system index i_k. (b) An output-only effect occupies the unprimed index o_k. (c) A two-leg map joins o_k to i_{k+1} between adjacent cores. (d) A tester T_y: the joint initial state ρ_{SA} (purple triangle) meets the first process input i_1, and there is no process output before that core; the open process ports are $i_1, o_1, i_2, o_2, i_3, o_3$. Subsequent joint maps $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$ span both rails, while ancilla-only operations $\mathcal{V}_k$ sit on the persistent A wire. The yellow dashed comb is the dual of Fig. 6(c), with open Liouville slots above the boundary. Note that the primed and unprimed legs appear in the opposite direction in a tester instrument to match the indices of the PT-MPO.

Figure 9 collects these cases and places them beside the persistent ancillary wire used by a tester.

The single-leg types are intentionally restrictive where the physics fixes their orientation. `StatePreparation` acts only on an input leg, and `TraceOut` acts only on an output leg by inserting the vectorised identity effect $\langle\langle\mathbb{I}|$. Likewise, `OpenInput` and `OpenOutput` are fixed to prime levels one and zero, respectively. `ObservableMeasurement` targets an output leg by default, but may be placed on an input leg with `leg_plev=1`; the latter is useful when an operator must act from the right in a correlation-function schedule. The two-leg constructors validate the neighbouring-time rule in Eq. (69) whenever concrete PT sites are supplied.

Table 2 gives the compact constructor dictionary needed for the remainder of the paper. These constructors store Hilbert-space states, Hamiltonians, observables, or maps without immediately producing a dense tensor. The corresponding Liouville-space `ITensor` is created only after a schedule is bound to a specific `ProcessTensor` in a contraction function. Several physicality distinctions in Table 2 are worth retaining. `identity_operation()` and unitary propagation are CPTP. A normalised density operator defines a deterministic preparation, and a positive effect bounded by the identity defines a probabilistic measurement outcome. By contrast, the general map $\rho \mapsto A\rho B$ is not necessarily completely positive, an arbitrary observable is not necessarily an effect, and a `CustomTwoLegInstrument` is not automatically checked for complete positivity or trace preservation. Even though these objects need not correspond to an operational instrument, they are still useful to compute physically relevant quantities such as multi-time correlation functions. A `ProductInstrument` is produced by multiplying an output-side and an input-side single-leg object; for a POVM measurement and a normalised preparation it realises the causal break in Eq. (65).

Constructor	PT legs	Role in a contraction
`state_preparation(`ρ`)`	input	Initial or replacement state
`observable_measurement(`O`)`	output/input	Effect or general observable insertion
`trace_out()`	output	Deterministic identity effect
`identity_operation()`	output–input	Identity channel between adjacent slots
`unitary_propagation(`H`, sites)`	output–input	Control generated by a Hamiltonian; $H(t)$ is also accepted
`left_right_operator(`A, B`)`	output–input	Linear map $\rho \mapsto A\rho B$
`left_action`, `right_action`	output–input	Convenient one-sided operator actions
`ProductInstrument`	separate pair	Output effect multiplied by an input preparation
`custom_twoleg_instrument(...)`	output–input	User-supplied dense Liouville map
`open_output()`, `open_input()`	one open leg	Keep a selected output or input unresolved
`open_inout()`	two open legs	Keep both legs unresolved and independent

Table 2: Principal lazy instrument constructors in `ProcessTensors.jl`. "PT legs" specifies which temporal indices the materialised object occupies.

`open_in, open_out, open_inout` constructors are bookkeeping operations rather than physical maps to keep track of which legs we wish to leave uncontracted by the end of a process evaluation. They materialise as the scalar tensor `ITensor(1.0)` during contraction, deliberately leaving the claimed PT indices uncontracted. They are different from `identity_operation()`. The former retains two independent indices, whereas the latter contracts them through an identity channel. Selective opening is important because `evaluate_process` returns a dense tensor whenever several system legs remain. If m Liouville legs of a d-dimensional system are open, the dense object contains d^{2m} complex entries. For a qubit, sixteen open legs already require 4^{16} complex numbers, or 64 GiB in `ComplexF64` before tensor overhead. Open-leg schedules should therefore expose only the temporal indices required by the calculation.

4.3 Assembling multi-time experiments using `InstrumentSeq`

An experiment rarely contains only one operation. `InstrumentSeq` is the lazy, time-indexed dictionary that records which instrument is inserted at every logical timestep. Its three fields are

$$\texttt{InstrumentSeq} = (\texttt{default}, \texttt{entries}, \texttt{nsteps}). \tag{71}$$

The `default` fills ordinary between-attachments for which no explicit action is given; `entries` is a sparse dictionary of overrides; and `nsteps` sets the upper core index within which the instruments will act. An attachment is stored by core index and leg role: $(k, \texttt{:input})$, $(k, \texttt{:output})$, or $(k, \texttt{:between})$. A preparation is $(1, \texttt{:input})$, a terminal effect is $(N, \texttt{:output})$, and a map between consecutive cores is $(k, \texttt{:between})$, which joins o_k to i_{k+1}. The input of core 1 has no default, because the initial system state is not part of the process tensor. An N-core process therefore materialises one preparation, $N-1$ between-attachments, and one terminal action.

The three-step process `pt_dense` from Sec. 3.5 can be probed by preparing an up spin, applying an additional control Hamiltonian during the second bridge, and reading out S^z at the final output:

Julia

```julia
psi0 = MPS(system_sites, ["Up"])
rho0 = to_dm(psi0)

H_control = OpSum()
H_control += 0.4, "Sy", 1

O_final = OpSum()
O_final += 1.0, "Sz", 1

seq = InstrumentSeq(
    default=identity_operation(),
    nsteps=pt_dense.nsteps,
)
seq += state_preparation(rho0), 1
seq += unitary_propagation(H_control, system.sites), 2
seq += observable_measurement(O_final), pt_dense.nsteps
```

Listing 14: A lazy instrument schedule for a controlled three-step experiment.

This incremental += syntax resembles the use of `OpSum` in `ITensorMPS.jl`, but each tuple assigns an instrument to a time slot rather than adding an operator term.

The schedule in Listing 14 corresponds to the scalar contraction

$$z = \langle\langle S^z|_{o_3} Q^{[3]} \mathcal{U}_c Q^{[2]} \mathcal{I} Q^{[1]} |\rho_0\rangle\rangle_{i_1}, \tag{72}$$

where the tensor contraction over the temporal memory indices χ_1, χ_2 and the joined system indices is implicit. The default identity supplies $\mathcal{I}$ at step 1, the explicit entry at step 2 supplies $\mathcal{U}_c$, and the terminal observable closes the last output. A callable control Hamiltonian on a between-attachment at anchor k is sampled at $(k - \frac{1}{2})\Delta t$, which is the midpoint of that propagation interval and is not the same statement as the attachment time $k\Delta t$. Replacing the terminal observable by `open_output()` turns the same history into a final reduced state; changing an intermediate entry to a measurement and preparation pair implements a causal break. The PT-MPO itself is unchanged in all three experiments.

During materialisation, consecutive single-leg output and input entries form their two-leg product. This allows a measurement at one side of a temporal cut and a preparation at the other to be written separately while retaining the product structure in Eq. (65). Conversely, incompatible placements are rejected before the costly contraction begins: an unpaired physical input action, a two-leg object at the terminal slot, or a timestep beyond `nsteps` produces an informative `ArgumentError` rather than a malformed tensor network.

4.4 Memory-bearing instruments: process testers

An `InstrumentSeq` describes a product schedule whose entries are chosen independently at different intervention times. A *process tester* is the more general multi-time experimental apparatus inserted into the open system legs of a process tensor [59, 60]. It may contain an ancillary quantum system A that is initialised once and retained throughout the experiment, allowing information acquired from the system at one intervention to influence controls applied at a later time.

A constructive realisation of such a tester is specified by

$$\mathbf{T} = \left(\rho_{SA},\ \{\mathcal{C}_k^{SA}\}_{k=0}^{N-1},\ \{\mathcal{V}_k^{A}\}_{k=0}^{N-1}\right), \tag{73}$$

where ρ_{SA} is the initial joint system–ancilla state,

$$\mathcal{C}_k^{SA} : \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_A) \longrightarrow \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_A) \tag{74}$$

is a joint system–ancilla operation, and

$$\mathcal{V}_k^{A} : \mathcal{B}(\mathcal{H}_A) \longrightarrow \mathcal{B}(\mathcal{H}_A) \tag{75}$$

acts only on the retained ancilla. These ingredients correspond directly to Fig. 9(d): ρ_{SA} enters at the right boundary, the maps $\mathcal{C}_k^{SA}$ span the system and ancillary rails, and the maps $\mathcal{V}_k^{A}$ act on the persistent ancillary wire as $\mathcal{I}_S \otimes \mathcal{V}_k^{A}$. Numerically, the former become two-site Liouville tensors and the latter one-site tensors, while the same ancillary index is contracted forward through the complete network. The system legs between successive joint maps remain available for contraction with the PT-MPO and therefore carry the opposite prime-level orientation to the matching process tensor legs.

Because the ancillary degree of freedom is not reset between interventions, the Choi tensor of a correlated tester generally does not factorise into time-local operations,

$$\mathbf{T} \neq \bigotimes_{k=0}^{N-1} \widehat{\mathcal{A}}_k. \tag{76}$$

The product form is recovered when the ancilla is trivial or is independently reset after every intervention. An outcome-valued tester is obtained by terminating the network with a POVM $\{F_y\}$,

$$F_y \geq 0, \qquad \sum_y F_y = \mathbb{I}_{SA}, \qquad p(y|\Upsilon) = \mathrm{Tr}\big[F_y\, \omega_{SA}(\Upsilon)\big], \tag{77}$$

where $\omega_{SA}(\Upsilon)$ is the final state produced after contracting the memory-bearing network with the process Υ. Absorbing this sequential realisation and its terminal POVM into Choi operators gives the process-POVM elements [59, 61]

$$T_y \geq 0, \qquad p(y|\Upsilon) = T_y \star \Upsilon, \qquad \sum_y T_y = T_{\mathrm{det}}, \tag{78}$$

where T_{det} is the causally normalised deterministic tester. Its sequential realisation uses CPTP maps $\mathcal{C}_k^{SA}$ and $\mathcal{V}_k^{A}$; an individual outcome branch may instead be CP and trace non-increasing, provided that the complete set of branches sums to a CPTP deterministic strategy.

`ProcessTensors.jl` represents Eq. (73) using two objects. A `Tester` stores the ancillary site and its initial state, whereas a `TesterSeq` stores the ancillary-only and joint actions applied at each timestep. System-only controls remain in the ordinary `InstrumentSeq`; in the current API, the initial state is assembled as $\rho_{SA} = \rho_S \otimes \rho_A$ from the two objects.

Julia

```julia
system_sites = siteinds("Qubit", 1)
tester_sites = siteinds("Qubit", 1)
rho_A0 = to_dm(MPS(tester_sites, ["1"]))
memory = tester(tester_sites, rho_A0)

U_SA = op("SWAP", only(system_sites), only(tester_sites))
U_A = op("X", only(tester_sites))
joint_U_SA = joint_unitary(U_SA, system_sites, tester_sites)
local_U_A = tester_unitary(U_A, tester_sites)

```

```
tester_seq = TesterSeq(nsteps=pt_dense.nsteps)
tester_seq += joint_U_SA, 1
tester_seq += local_U_A, 2
tester_seq += joint_U_SA, 3
```

Listing 15: A single-qubit tester with reusable joint and local unitaries.

The site ordering of `U_SA` is system then tester, matching `vcat(system_sites, tester_sites)`. The joint unitary at steps 1 and 3 couples the system and ancilla, while the local unitary at step 2 acts only on the retained ancilla memory. Unspecified timesteps use `tester_identity()`. The current implementation supports one tester site and joint actions on one system site and one tester site. Tester actions are kept in `TesterSeq` and cannot be added to an `InstrumentSeq`, preventing the two schedules from being mixed.

4.5 Evaluating and evolving a process tensor

The objects introduced above are completed by `evaluate_process`. Given a PT-MPO Υ and a schedule $\mathbf{A}$, the function validates the claimed temporal legs, materialises each lazy instrument on the corresponding PT sites, optionally compiles the tester memory wire, and contracts the resulting network chronologically. Abstractly,

$$R_{\mathbf{A}} = \mathrm{Contr}\big[\Upsilon, \mathcal{P}_{\rho_0}, \mathcal{A}_1, \ldots, \mathcal{A}_{N-1}, \mathcal{M}_N\big], \tag{79}$$

where the nature of $R_{\mathbf{A}}$ is determined by the system indices that the schedule leaves unresolved. Figure 10 shows the three return cases and the related sequence of snapshot contractions used by `evolve`.

Open system legs	Julia return type	Interpretation
0	`ComplexF64`	Probability, trace, expectation value, or correlation scalar
1	`MPS{Liouville}`	Reduced state-like Liouville vector
2 or more	`ITensor`	Dense partially contracted multi-time object

Table 3: Return type of one `evaluate_process` call. A batch of fully contracted schedules returns one `Vector{ComplexF64}`.

A result with several open legs can be interpreted mathematically as a reduced or partially contracted process tensor, but its present Julia type is a dense `ITensor`, not a `ProcessTensor`. This distinction matters because the returned object no longer carries the PT-MPO metadata or temporal compression of the original process. A fully contracted result is a probability only when the terminal and intermediate insertions form a physical instrument; with a general observable or left–right action, the same scalar may instead be an expectation value or correlation function.

The function `open_leg_info` predicts which input and output times remain unresolved before a contraction is attempted. In the following schedule, `open_inout()` at step 2 leaves o_2 and i_3 independent while the initial preparation and final trace close the remaining boundary legs:

Julia

```julia
seq_open = InstrumentSeq(
    default=identity_operation(),
    nsteps=pt_dense.nsteps,
)
add!(seq_open, state_preparation(rho0), 1)
add!(seq_open, open_inout(), 2)
add!(seq_open, trace_out(), pt_dense.nsteps)

info = open_leg_info(pt_dense, seq_open)
reduced = evaluate_process(pt_dense, seq_open; progress=false)

println("open input tsteps: ", info.open_in)
println("open output tsteps: ", info.open_out)
println("expected open legs: ", info.n_open_expected)
println("open dimensions: ", info.open_dims)
println("return type: ", typeof(reduced))
```

Terminal

```
open input tsteps: [3]
open output tsteps: [2]
expected open legs: 2
open dimensions: [4, 4]
return type: ITensor
```

Listing 16: Inspecting the open legs before constructing a dense reduced process.

This diagnostic turns the d^{2m} open-leg scaling into an immediate memory estimate. For example, the dimensions in Listing 16 predict a 4×4 dense object before the PT contraction begins. The helper also reports missing input or output coverage, allowing malformed schedules to be detected independently of the eventual return type.

The essential software advantage is now visible. Once `pt_dense` or an ACE-compressed PT has been constructed, different sequences can be evaluated without rebuilding the environmental influence. A terminal trace tests normalisation, a terminal observable gives an expectation value, a selected positive effect gives an outcome probability, an intermediate measurement–preparation pair gives a conditional trajectory, and selected open legs expose a reduced map. The same mechanism also supports the multi-time left–right insertions used by correlation functions.

The tester of Sec. 4.4 is contracted by the same function. Recalling Listing 15, `memory` carries the ancillary initial state and `tester_seq` stores the joint and local unitaries; `seq` remains the system schedule of Listing 14. Passing both objects into `evaluate_process` compiles the persistent A wire of Fig. 9(d) and traces it out after the final interval, so the return type is still fixed by the open *system* legs:

Julia

```julia
result = evaluate_process(
    pt_dense,
    seq;
    tester=memory,
    tester_seq=tester_seq,
```

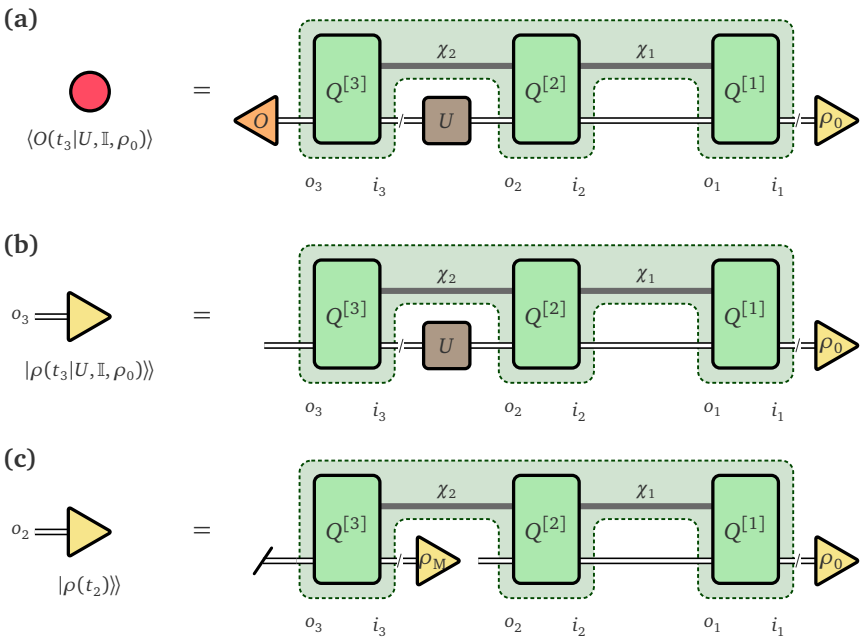


Figure 10: Evaluation and evolution of the instrument sequence in Listing 14. The dashed comb is the process tensor, while $|\rho_0\rangle\rangle$, $\mathbb{I}$, U, and $\langle\langle O|$ are the instruments that sit on the open system legs. (a) `evaluate_process` with an expectation of an observable: every system leg is closed and the process returns the conditional scalar $\langle O(t_3|U,\mathbb{I},\rho_0)\rangle$. (b) The same history with `open_output()` at the last slot: o_3 remains unresolved and the process returns the conditional Liouville state $|\rho(t_3|U,\mathbb{I},\rho_0)\rangle\rangle$. (c) An `evolve` snapshot at t_2: past controls are retained, the output o_2 is left open, and the unused future core $Q^{[3]}$ is closed by a trace-out on o_3 and a preparation $|\rho_M\rangle\rangle^{i_3} = |\mathbb{I}/d\rangle\rangle^{i_3}$.

```
    progress=false,
)
```

Listing 17: Evaluating the process tensor against the tester of Listing 15.

For the common special case in which the desired output is the reduced system state at every stored timestep, repeatedly assembling schedules by hand would be unnecessary. The `evolve` interface performs these contractions and returns both Liouville- and Hilbert-space trajectories. A custom `InstrumentSeq` may be supplied in place of `rho0` alone when the trajectory includes control operations:

Julia

```
trajectory = evolve(pt_dense, seq; progress=false)

times = trajectory.times
rho_liouville = trajectory.states_liouville
rho_hilbert = trajectory.states_hilbert
```

Listing 18: Reduced dynamics from the process tensor and the instrument sequence of Listing 14.

The returned arrays contain the initial state together with one snapshot after every stored

PT core, so an N-core process yields $N+1$ states. For the three-step example the array index $j = 1,2,3,4$ labels the physical times $0, \Delta t, 2\Delta t, 3\Delta t$. Later instruments in `seq` cannot affect an earlier reduced state since they are protected by the causality (no-signalling-from-the-future) property of the process tensor, Eq. (40) in Sec. 3.1. With a tester present, `states_liouville` and `states_hilbert` still contain the reduced system states; `tester_states_{liouvil` are returned by default, and the joint fields `joint_states_{liouville,hilbert}` can be requested with `return_joint=true`.

Figure 10 depicts the schedule for one snapshot. The past preparation and controls are retained, the required output is left open, and every later core is discarded through the disconnected CPTP closure

$$\langle\langle \mathbb{I}|_{o_j} \otimes |\mathbb{I}/d\rangle\rangle^{i_{j+1}}. \tag{80}$$

The reference option `contraction=:evaluate` constructs precisely this full schedule and calls `evaluate_process` once per snapshot, giving an $O(N^2)$ contraction in the number of PT cores. The default `contraction=:closures` evaluates the same trajectory with one backward sweep of future closures followed by one forward prefix contraction, reducing the timestep scaling to $O(N)$. For a compressed PT, these backward closure tensors are essential: a temporal bond χ_k is an effective memory basis and cannot in general be terminated by inserting a bare bath trace vector.

5 Examples of process tensor workflows

With the theory and programming interface of the previous sections in place, we now turn to physical examples. We demonstrate how the reduced system evolves, what the probability of a multi-round experimental record is, how operators at different times are correlated, and what changes when the intervention itself carries a persistent quantum memory. The following examples reproduce or adapt established open-system and quantum-information protocols; each is obtained by changing only the instruments contracted with a common `ProcessTensor`. We give a brief physical motivation and the core code that produces the figures; complete walkthroughs are in the package documentation.

5.1 Reduced dynamics of a central spin

A first use of a process tensor is the familiar "recover the reduced dynamics of a system after its environment has been eliminated". We borrow, for this purpose, the fully polarised central-spin benchmark used by Cygorek *et al.* to demonstrate ACE for a non-Gaussian spin environment [20]. A central spin-1/2, $\mathbf{S}$, interacts isotropically with N noninteracting bath spins $\mathbf{s}_k$,

$$H_S = H_E = 0, \qquad H_{SE} = \sum_{k=1}^{N} \frac{J_k}{\hbar^2}\, \mathbf{S}\cdot\mathbf{s}_k, \qquad J_k = \frac{J}{N}. \tag{81}$$

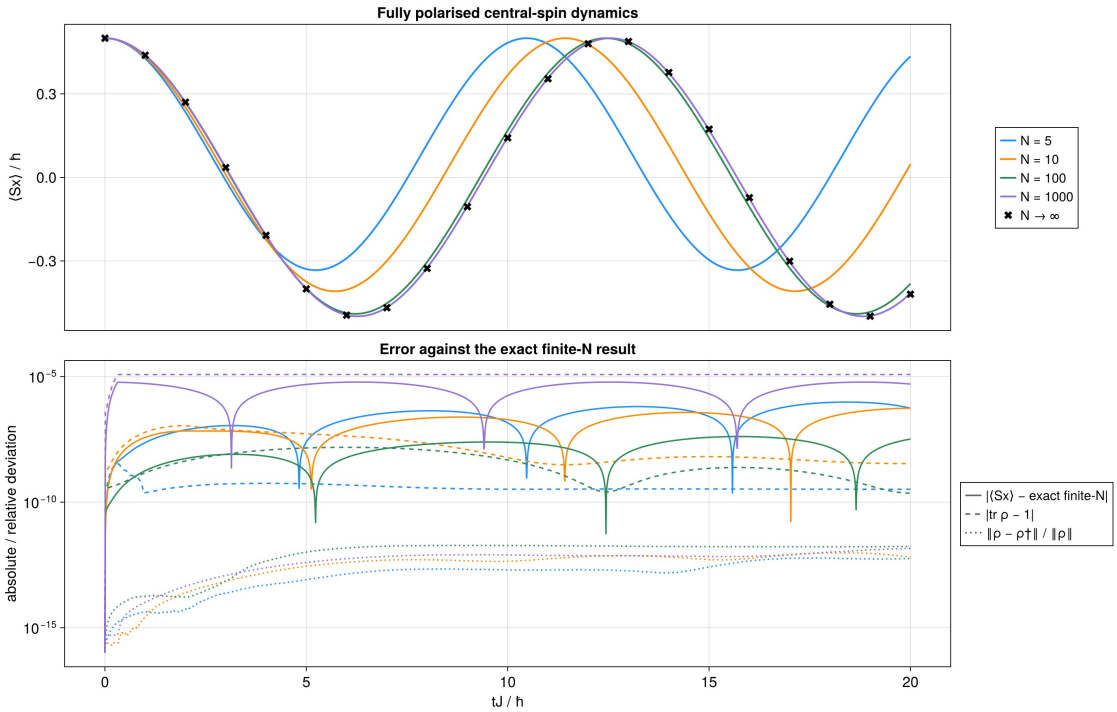


Figure 11: **Upper panel: fully polarised central-spin dynamics.** Reduced dynamics of the system spin coupled to a finite number of bath spins. Finite baths of the model in Eq. (81) approach the analytical $N \to \infty$ precession in Eq. (82) as N increases. **Lower panel: deviation from the exact finite-N result.** The solid curves are $|\langle S_x\rangle - \langle S_x\rangle_{\text{exact}}|$, and the other curves are $|\mathrm{Tr}\,\rho - 1|$ and $\|\rho - \rho^\dagger\|/\|\rho\|$.

The system is initially polarised along $+x$, while every bath spin begins along $+z$. Keeping $\sum_k J_k = J$ fixed prevents the interaction energy scale from growing with the number of modes. The system and bath spins have no free Hamiltonian that leads to a unitary evolution in their own subspace. Since neither subsystem has a free Hamiltonian, all non-trivial dynamics are generated by the system–bath interaction and depend on the initially polarised bath state. For finite N, back-action from the central spin generates correlations among bath spins even though they do not interact directly. These mediated correlations produce finite-size deviations from the collective precession. However, in the limit $N \to \infty$, the bath experiences vanishing back-action and produces a static effective magnetic field, giving

$$\frac{\langle S_x(t)\rangle}{\hbar} \longrightarrow \frac{1}{2}\cos\left(\frac{Jt}{2\hbar}\right). \tag{82}$$

Each bath spin is supplied to `ACE()` as an independent `SpinMode`. Once their joint influence has been compressed into a PT-MPO, the microscopic bath is no longer needed for the reduced calculation.

Julia

```julia
bath_sites = siteinds("S=1/2", N)
bath_sites_L = liouv_sites(bath_sites)

modes = SpinMode[]
for k in 1:N
    rho_k = to_dm(MPS([bath_sites[k]], ["Up"]))
    rho_k_L = to_liouville(rho_k; sites=[bath_sites_L[k]])
```

```
    coupling = OpSum()
    coupling += J / N, "Sx", 1, "Sx", 2
    coupling += J / N, "Sy", 1, "Sy", 2
    coupling += J / N, "Sz", 1, "Sz", 2
    push!(
        modes,
        spin_mode([bath_sites_L[k]], OpSum(), rho_k_L;
        ↪ coupling=coupling),
    )
end
bath = spin_bath(modes)

process_tensor = build_process_tensor(
    system; method=ACE(cutoff=1e-10),
    environment=bath, dt=0.01, nsteps=nsteps,
)

trajectory = evolve(process_tensor, rho_S0)
Sx_t = [real(tr(Sx * density_matrix(rho)))
        for rho in trajectory.states_hilbert]
```

Listing 19: Independent bath spins assembled into an ACE process tensor.

5.2 Repeated Ramsey measurements with active reset

Reduced dynamics observes the system without intervening. A process tensor also predicts the joint statistics of an experiment in which measurements and controls are applied repeatedly. We illustrate this using Ramsey interferometry, introduced originally as a separated-oscillatory-field spectroscopy protocol [62], and now ubiquitous in qubit characterisation. This choice is also motivated by recent studies of temporally correlated circuit noise generated by multi-mode spin–boson environments [63,64]. The qubit is coupled through Z to a thermal bosonic environment described by the pure-dephasing spin–boson model,

$$H_E = \sum_k \omega_k b_k^\dagger b_k, \qquad H_{SE} = Z \sum_k \sqrt{J(\omega_k)}(b_k + b_k^\dagger), \qquad J(\omega_k) = 2\alpha\omega_k e^{-\omega_k/\omega_c}, \tag{83}$$

which is a standard model of structured dephasing noise [65]. The continuum is discretised into thermal oscillator modes and compressed once with `ACE()`.

The experimental protocol performed on the qubit is as follows: the qubit is initially prepared in $\rho_r = |+\rangle_x\langle+|$, evolves for a fixed Ramsey time, and is read out with the POVM

$$E_x^{(\eta)} = \frac{1}{2}\big(\mathbb{I} + x\eta\sigma_x\big), \qquad x \in \{-1,+1\}, \qquad 0 \le \eta \le 1. \tag{84}$$

Here, η is the effective measurement visibility (sharpness), quantifying how reliably the Ramsey readout distinguishes the two X-basis outcomes. A POVM element fixes the outcome probability but not the state supplied to the following round. We therefore combine the readout with an outcome-independent active reset and use the causal-break operation

$$\mathcal{A}_x^{\mathrm{MR}}(\rho) = \mathrm{Tr}\big[E_x^{(\eta)}\rho\big]\rho_r, \qquad = |\rho_r\rangle\!\rangle[\mathcal{A}_x^{\mathrm{MR}}]\langle\!\langle E_x^{(\eta)}|. \tag{85}$$

Each outcome map is CP and trace-non-increasing, while their sum is the CPTP replacement channel

$$\sum_x \mathcal{A}_x^{\mathrm{MR}}(\rho) = \mathrm{Tr}(\rho)\rho_r. \tag{86}$$

Measure-and-reprepare causal breaks are central to the operational definition and experimental reconstruction of non-Markovian processes [6, 66, 67]. A causal break introduced in the system will ensure no information of the qubit's previous states before this break affects its future dynamics through the qubit itself. If any correlation is observed in the following dynamics, it should come from the process tensor of the environment.

`ProcessTensors.jl` does not yet provide a dedicated POVM container in v0.3.0. Each outcome map in Eq. (85) is instead assembled from two single-leg instruments: an output-leg `observable_measurement` of $E_x^{(\eta)}$ and an input-leg `state_preparation` of ρ_r. The package overloads `*` for this purpose. Provided the two instruments have opposite prime levels—one with plev $= 0$ and one with plev $= 1$—their product is a `ProductInstrument`, a two-leg object inserted at a single evolve slot of an `InstrumentSeq`. If we perform this POVM measurement and reset three times, we obtain the joint probability $p(x_1, x_2, x_3)$ of the record $|x_1\rangle_x, |x_2\rangle_x, |x_3\rangle_x$.

Julia

```julia
eta = 0.90
rho_r = to_dm(MPS(system_sites, ["+"]))

effects = Dict(x => (OpSum() + (0.5, "Id", 1) + (0.5 * x * eta, "X", 1)) for x in (-1, 1))
instruments = Dict(
    x => observable_measurement(effects[x]) * state_preparation(rho_r)
    for x in (-1, 1)
)

for record in Iterators.product((-1, 1), (-1, 1), (-1, 1))
    seq = default_schedule(process_tensor)
    seq += state_preparation(rho_r), 1
    for (k, x) in zip(readout_steps, record)
        seq += instruments[x], k
    end
    seq += trace_out(), process_tensor.nsteps
    probabilities[record] = real(evaluate_process(process_tensor, seq))
end
```

Listing 20: POVM-and-reset `ProductInstruments` for three Ramsey readouts.

Evaluating the process tensor for each of the above records, the returned scalar is the joint probability

$$p(x_1, x_2, x_3) = \mathrm{Tr}\Big[\mathcal{T}\big[\mathcal{A}^{\mathrm{MR}}_{x_3}, \mathcal{A}^{\mathrm{MR}}_{x_2}, \mathcal{A}^{\mathrm{MR}}_{x_1}, \mathcal{P}_{\rho_r}\big]\Big]. \tag{87}$$

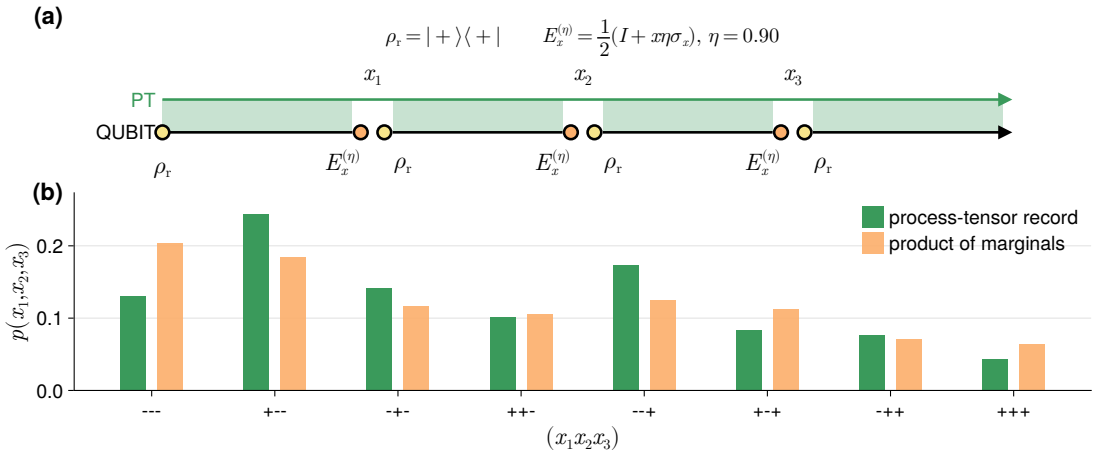


Figure 12: **(a) Timeline of the Ramsey readout protocol.** Three repeated readouts of the thermally dephasing qubit in Eq. (83), each followed by the same active reset, Eq. (85). The time exposed to the environment between each prepare and measure is the same for all three readouts. The green fill between the PT and QUBIT line depicts constant interaction between them. **(b) Outcome-record probabilities $p(x_1, x_2, x_3)$.** For some three-shot records the process tensor probabilities do not coincide with the products of the single-round marginals, indicating correlations between successive readouts.

The eight probabilities in Fig. 12 are non-negative and sum to unity, as required by Eq. (86). For a memoryless process, the active reset makes the three identically timed rounds independent, so that the joint distribution factorises as $p(x_1, x_2, x_3) = \prod_j p(x_j)$. Deviations from the product of the measured single-round marginals therefore reveal information retained by the environment across causal breaks. Thus, this is an operational signature of non-Markovianity.

5.3 Multi-time correlations in a spin environment

Correlation functions at multiple times govern response theory, noise spectroscopy, and multidimensional spectroscopy. In a non-Markovian problem they cannot, in general, be reconstructed from the reduced trajectory alone because the first operator insertion changes the subsequent joint system–environment evolution. PT-MPO calculations of such correlations have been developed for strongly coupled spin chains and structured baths [68], and have since been used in process tensor approaches to nonlinear spectroscopy [69]. We take a minimal two-spin analogue so that every result can also easily be checked against direct joint evolution,

$$H_S = S_x, \qquad H_E = S_x, \qquad H_{SE} = S_z \otimes S_z, \tag{88}$$

with the system initially in $|\downarrow\rangle$ and the environment in $|\uparrow\rangle$.

For $t_2 \geq t_1$, the ordered correlation is

$$C_{AB}(t_2, t_1) = \mathrm{Tr}_{SE}\big[(A \otimes \mathbb{I}_E)U_{2:1}(B \otimes \mathbb{I}_E)U_{1:0}\rho_{SE}(0)U_{1:0}^\dagger U_{2:1}^\dagger\big]. \tag{89}$$

Thus B is inserted from the left at t_1, the resulting operator is propagated to t_2, and A closes the final leg. The order is reversed when $t_1 > t_2$, while coincident insertions are combined on the same temporal leg. These cases are encoded by `two_time_correlation_seq`, so a complete grid reuses one process tensor:

Julia

```
Czz = [
    evaluate_process(
        process_tensor,
        two_time_correlation_seq(
            process_tensor, (sigma_z, n2), (sigma_z, n1);
            rho0=rho_S0,
        ),
    )
    for n1 in time_indices, n2 in time_indices
]
```

Listing 21: Two-time correlation grid from one stored process tensor.

For $t_2 > t_1$, the helper translates Eq. (89) into instruments: $\rho_S(0)$ is prepared at the first input, B is inserted as a left action at t_1, A is inserted as a left action at t_2, and a terminal `trace_out` realises the partial trace. The arguments of `two_time_correlation_seq` are snapshot indices in `1:(pt.nsteps+1)`. Snapshot 1 is the initial state, so a correlator that includes t_0 starts at index 1, and $C_{zz}(t_3, t_1)$ uses snapshots 4 and 2. On a five-step process with $(n_2, n_1) = (4, 2)$, the resulting schedule is

Julia

```
seq = two_time_correlation_seq(
    process_tensor, (sigma_z, 4), (sigma_z, 2);
    rho0=rho_S0,
)
println(seq)
```

Terminal

```
InstrumentSeq(default=IdentityOperation, nsteps=5, 5 explicit entries)
  (1, :between) => LeftRightOperator{MPO{Hilbert, Nothing}, MPO{Hilbert,
  ↪ Nothing}}
  (1, :input) => StatePreparation{MPO{Hilbert, Nothing}}
  (3, :between) => LeftRightOperator{MPO{Hilbert, Nothing}, MPO{Hilbert,
  ↪ Nothing}}
  (4, :between) => IdentityOperation
  (5, :output) => TraceOut
```

Listing 22: Instrument sequence for $C_{zz}(t_3, t_1)$ on a five-step process.

The entry $(1, \texttt{:input})$ prepares $\rho_S(0)$. The left action at $(1, \texttt{:between})$ is $B = \sigma_z$ at t_1, and the left action at $(3, \texttt{:between})$ is $A = \sigma_z$ at t_3. The schedule also records an identity at $(4, \texttt{:between})$ before the terminal trace at $(5, \texttt{:output})$. If $t_1 > t_2$, the early insertion becomes a right action; if $t_1 = t_2$, the two operators are multiplied on the same temporal leg.

Several expectations follow without solving the dynamics. Hermiticity and $\sigma_z^2 = \mathbb{I}$ give

$$C_{zz}(t, t) = 1, \qquad C_{zz}(t_2, t_1)^* = C_{zz}(t_1, t_2), \tag{90}$$

so the real and imaginary parts are respectively symmetric and antisymmetric under $t_1 \leftrightarrow t_2$. The equal-time cross-correlations instead follow from

$$\sigma_z \sigma_x = i\sigma_y, \qquad \sigma_x \sigma_y = i\sigma_z. \tag{91}$$

Consequently, the real parts of the corresponding equal-time correlations vanish. The numerical grids in Fig. 13 reproduce these correlation properties faithfully.

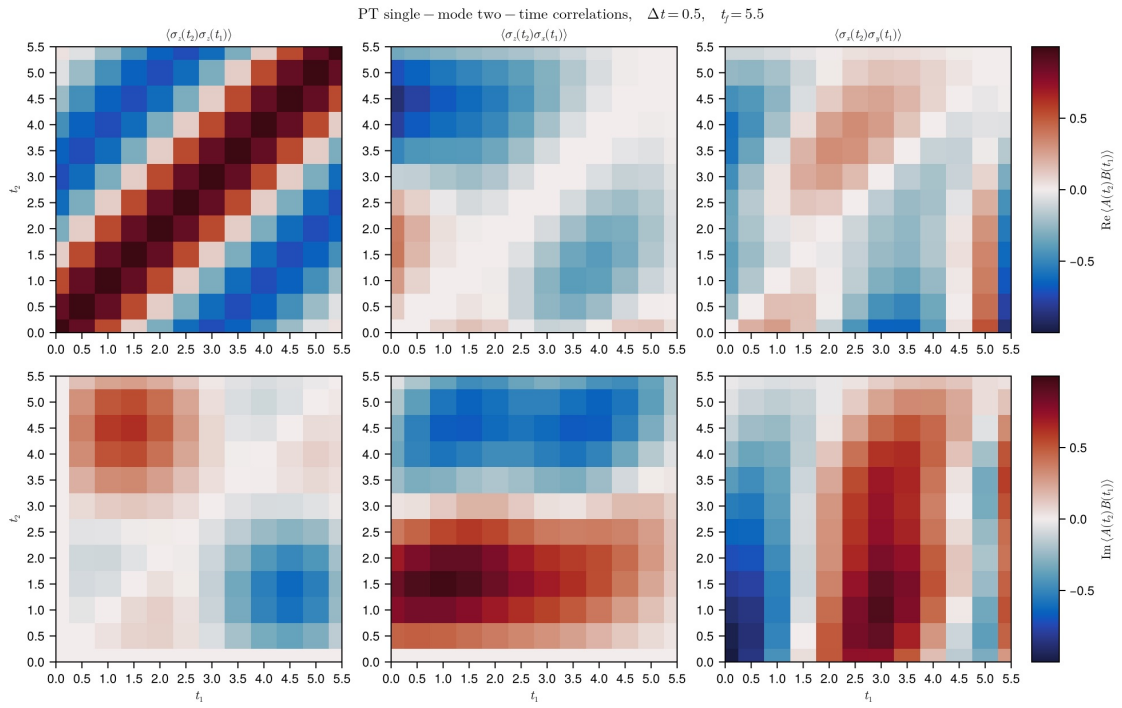


Figure 13: **Two-time spin correlations from a reused process tensor.** The upper panels show the real parts of C_{zz}, C_{zx}, and C_{xy} for the two-spin model in Eq. (88); the lower panels show the corresponding imaginary parts. The autocorrelation has a unit real diagonal and the conjugate symmetry of Eq. (90); the cross-correlation diagonals follow Eq. (91). Every pixel is obtained by changing only the two operator-insertion times of the same stored process tensor.

5.4 Memory-assisted control with a quantum tester

The previous experiments repeatedly removed the system's local memory. We finally consider the opposite setting, in which the intervention deliberately retains a quantum memory. Quantum testers and combs were introduced to describe the most general probing strategies for channels used over several times, including protocols in which an ancilla persists between uses [59, 70]. Such memory-assisted strategies are particularly natural for correlated circuit noise, whose implications for randomized benchmarking have been analysed in Refs. [63, 64].

We return to the thermal spin–boson process of Eq. (83), but attach an isolated ancillary qubit A to the noisy processor qubit S. The protocol is as follows: first SWAP the system and ancilla to store the logical state in A, then apply a Z_A gate only on the ancilla while it is parked away from the bath, and then do the second SWAP to retrieve it. The persistent ancillary wire turns this sequence into a temporally correlated tester.

Julia

```
memory = tester(ancilla_sites, rho_A0)

U_SA = op("SWAP", only(system_sites), only(ancilla_sites))
U_A = op("Z", only(ancilla_sites))
```

```
joint_U_SA = joint_unitary(U_SA, system_sites, ancilla_sites)
local_U_A = tester_unitary(U_A, ancilla_sites)

controls = TesterSeq(nsteps=process_tensor.nsteps)
controls += joint_U_SA, store_step
controls += local_U_A, phase_step
controls += joint_U_SA, retrieve_step

trajectory = evolve(
    process_tensor, rho_S0;
    tester=memory, tester_seq=controls, return_joint=true,
)
```

Listing 23: SWAP–phase–SWAP tester sequence for memory-assisted control.

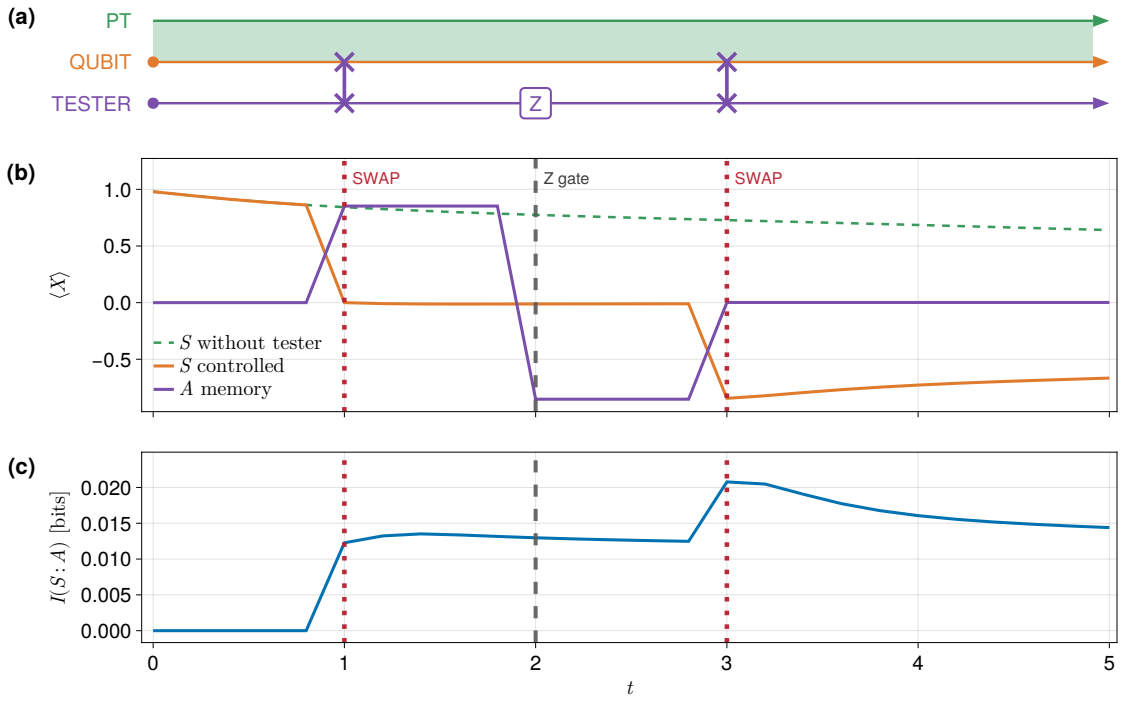


Figure 14: **(a) Schematic of the SWAP–phase–SWAP protocol.** The process tensor acts on the processor qubit; two SWAPs exchange that qubit with the tester, and a Z gate acts on the tester while the logical state is parked away from the bath. **(b) Transverse magnetisation.** The first SWAP transfers $\langle X \rangle$ from the noisy processor S to the isolated ancilla A, the Z gate reverses its sign, and the second SWAP returns it; without a tester, S continues to dephase. **(c) System–ancilla mutual information.** The first SWAP parks the logical state in A while the environment retains the processor history; later evolution and the retrieval SWAP convert part of that history into $I(S{:}A) \approx 0.02$ bits. The small magnitude reflects the weak-coupling parameters used here.

The output of evolving a PT-MPO with a tester contains the reduced states $\rho_S(t)$ and $\rho_A(t)$ by default. The joint trajectory $\rho_{SA}(t)$ is obtained by setting `return_joint=true`. We quan-

tify the total correlations of that joint state by the quantum mutual information

$$I(S{:}A) = S(\rho_S) + S(\rho_A) - S(\rho_{SA}), \qquad S(\rho) = -\mathrm{Tr}\big(\rho \log_2 \rho\big). \tag{92}$$

The panels of Fig. 14 follow this protocol. With no tester, the processor dephases under the thermal bath. The first SWAP parks the logical state in the isolated ancilla, while the environment retains correlations with the history of the processor. Subsequent evolution and the final SWAP convert part of that environmental history into system–ancilla correlations, so the observed $I(S{:}A)$ is a signature of the process tensor rather than of the joint gates alone. Its peak value of about 0.02 bits is small because the spin–boson coupling used here is weak. The two memories remain distinct: environmental memory is compressed into the PT-MPO, whereas the experimenter's controllable memory is carried by the tester ancilla.

Taken together, these workflows move from an unobserved reduced trajectory to outcome-resolved causal breaks, operator insertions at two times, and finally a coherent intervention with its own memory. Their common implementation is the central design choice of `ProcessTensors.jl`: the expensive environment is encoded once, while the experiment is changed by replacing the instruments attached to its temporal legs.

6 Benchmarks of the ACE algorithm

The preceding sections established the theory and representation of a process tensor, and showed how the same `ProcessTensor` can be interrogated with different instruments and testers. We now map the practical performance of its ACE constructor: how time-discretisation errors scale, how the timestep and cutoff affect PT-MPO compressibility, and how the implementation performs against the original C++ ACE toolkit.

All data are generated by scripts in the `benchmark/` directory in the main repository. Unless stated otherwise, both the system and mode-combination steps use second-order Trotterisation. We use `Paanini`, an AMD Ryzen Threadripper 7980X workstation with 64 physical cores, 128 hardware threads, 256 GiB RAM, and running on Ubuntu 22.04.5 to run all of our benchmarks in this section.

6.1 Correctness and discretisation errors

Our validation model is a single spin coupled to four independent bath spins,

$$H = S_x^{(S)} + \sum_{m=1}^{4} \left[\omega_m S_x^{(B_m)} + g_m S_z^{(B_m)} S_z^{(S)}\right], \quad \omega_m = 0.5 + 0.1m, \quad g_m = 0.2 + 0.3m, \tag{93}$$

with $\hbar = 1$, $S_\alpha = \sigma_\alpha/2$, and all spins initially in $|\uparrow\rangle$. The joint Hilbert space has dimension $2^5 = 32$, allowing direct diagonalisation without timestep discretisation to provide the reference trajectory.

Figure 15 compares direct system–environment evolution with uncompressed `Dense()` and compressed `ACE()` process tensors for $\Delta t = 0.20$, 0.10, and 0.05. The plotted observable is the transverse system-spin polarisation $\langle\sigma_y(t)\rangle$, evaluated from the reduced state as $\mathrm{Tr}[\sigma_y \rho_S(t)]$. The compressed tensors are constructed with `ACE(cutoff=1e-12, compression=:canonzip)`. We define

$$\varepsilon_{\sigma_y}(t) = \left|\langle\sigma_y(t)\rangle_{\text{method}} - \langle\sigma_y(t)\rangle_{\text{ED}}\right|. \tag{94}$$

In panel (a), the `Dense()` and `ACE()` trajectories lie on top of one another and closely follow the discretisation-free ED result. Panel (b) resolves the remaining timestep error: the error curves shift systematically downward as Δt is reduced from 0.20 to 0.10 and then to 0.05. The

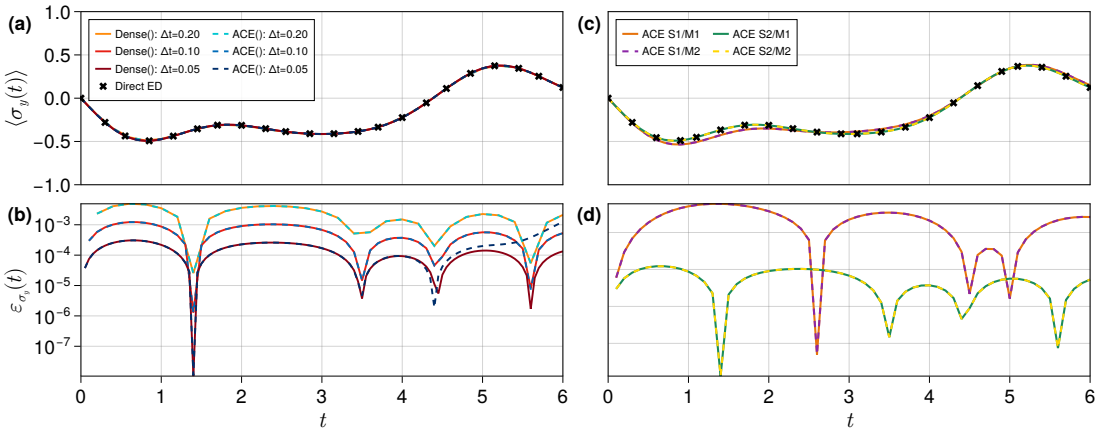


Figure 15: Correctness and propagation errors for the four-mode spin bath. Panel (a) shows the system-spin observable $\langle\sigma_y(t)\rangle$ from direct ED, uncompressed `Dense()`, and `ACE()` for three different Δt values; panel (b) shows their absolute errors relative to ED. Panels (c,d) compare first- and second-order system and mode-combination splittings at fixed Δt.

near coincidence of the `Dense()` and `ACE()` errors at each timestep shows that compression at the stated cutoff adds no visible error on this scale; the observed convergence towards ED is instead controlled by the time discretisation and propagation splitting.

At fixed $\Delta t = 0.10$, panels (c) and (d) separate the two Trotter choices made during an ACE construction. The labels S1 and S2 refer to the system–environment splitting in Eq. (54): S1 applies the environmental and free-system maps in an asymmetric first-order sequence, whereas S2 places half of the free-system evolution on either side of the environmental map to give the symmetric second-order sequence. The labels M1 and M2 make the analogous distinction when the independently propagated mode influences of Eq. (55) are combined: M1 joins them in a first-order ordered product, while M2 uses a symmetric second-order composition before the SVD compression in Eq. (56). For this particular four-spin bath the mode generators commute, so changing M1 to M2 leaves the result unchanged; the visible improvement comes from replacing S1 by S2.

In general, the error in an observable will stem from four sources:

$$\varepsilon_{\text{obs}} \lesssim \varepsilon_{\Delta t} + \varepsilon_{\text{sys}} + \varepsilon_{\text{mode}} + \varepsilon_{\text{SVD}}. \tag{95}$$

Here $\varepsilon_{\Delta t}$ is the finite-timestep propagation error, ε_{sys} and $\varepsilon_{\text{mode}}$ arise from the system–environment and intermode Trotter splittings, and ε_{SVD} is introduced by SVD truncation. For the commuting mode generators used here, $\varepsilon_{\text{mode}} = 0$; the observed deviation from ED is therefore set by the remaining contributions, dominated in this converged calculation by the timestep.

6.2 Temporal bond dimensions and compressibility

For the temporal-bond study we use the unpolarised central-spin model

$$H = \frac{J}{N}\sum_{k=1}^{N}(S_x s_k^x + S_y s_k^y + S_z s_k^z), \qquad H_S = H_{B_k} = 0, \tag{96}$$

with the system initially along $+x$ and independently sampled pure bath-spin states, uniform on the Bloch sphere. A single archived `Random.Xoshiro(20260905)` realisation defines the bath orientations for every run. The sweep uses $N = 50$, $T = 21$, $\Delta t \in \{1.5, 0.5, 0.2, 0.1, 0.05\}$, and $\epsilon_{\text{SVD}} \in \{10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}\}$.

Each internal bond χ_k of a PT-MPO crosses the temporal cut at $t_k = k\Delta t$. Its dimension $D_k = \dim(\chi_k)$ counts the effective memory states needed to transmit the influence of the history before that cut to interventions made afterwards. Inspecting D_k along the time direction therefore shows where the environmental history is easiest or hardest to compress, rather than reducing the whole process to a single cost estimate.

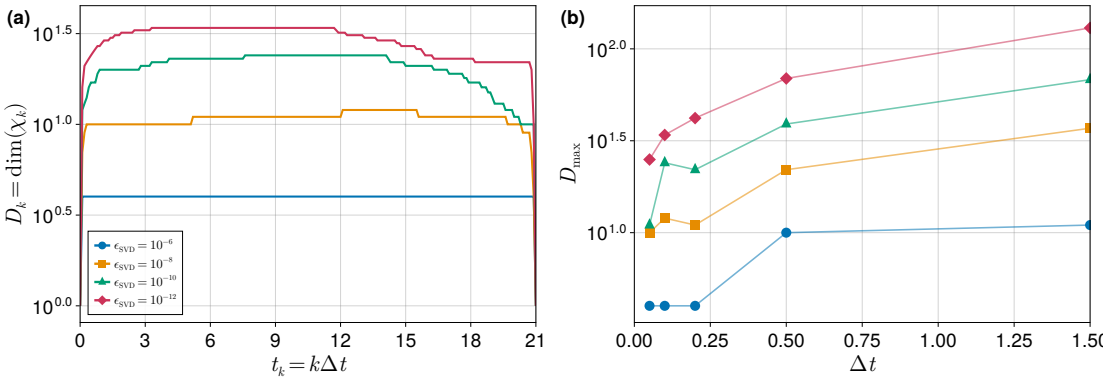


Figure 16: Temporal compressibility of the unpolarised central-spin process tensor. (a) Internal bond dimensions along the temporal MPO for several relative cutoffs. (b) Maximum internal bond dimension versus timestep and cutoff. Open symbols indicate runs limited by `maxdim`; the bath realisation is fixed across all points.

Figure 16 applies this diagnostic to the unpolarised central-spin model in Eq. (96). Panel (a) follows D_k across all internal time legs. Tightening the relative SVD cutoff retains weaker singular directions at every cut, so the temporal memory spaces become larger and the bond profile rises. This is the expected accuracy–cost trade-off: a tighter cutoff preserves more channels through which past system–bath correlations can affect the future.

Panel (b) condenses each profile to

$$D_{\max} = \max_{1 \le k \le n_{\mathrm{steps}}-1} D_k, \tag{97}$$

excluding the trivial boundary bonds, and compares this maximum across timesteps and cutoffs. At a fixed cutoff, the observed $D_{\max}$ increases with Δt. Over a longer interval the system and bath undergo more non-trivial joint dynamics before the next temporal leg is exposed; the resulting one-step influence can explore a larger bath operator subspace and therefore requires more effective memory states to represent. For smaller Δt, each individual step is closer to the identity and typically occupies a simpler subspace, although more PT-MPO cores are needed to reach the same final time.

6.3 Compression schedules and accuracy–cost trade-offs

The central-spin scaling below uses the model of Eq. (96), while the cutoff-versus-error comparison additionally uses a heterogeneous eight-spin environment. The package currently provides `:zipup_cpp` and `:canonzip` compression schedules. The former follows the C++ ACE join order: truncation is applied during the forward join and the resulting chain is swept backward. The latter forms the complete mode join, canonicalises towards the final time, and performs a relative SVD sweep from right to left. Both use the same cutoff and `maxdim`; only the schedule differs.

Figure 17 compares these schedules. Panels (a) and (b) show that `:zipup_cpp` is consistently faster and allocates less memory than `:canonzip` as the spin bath grows. This saving

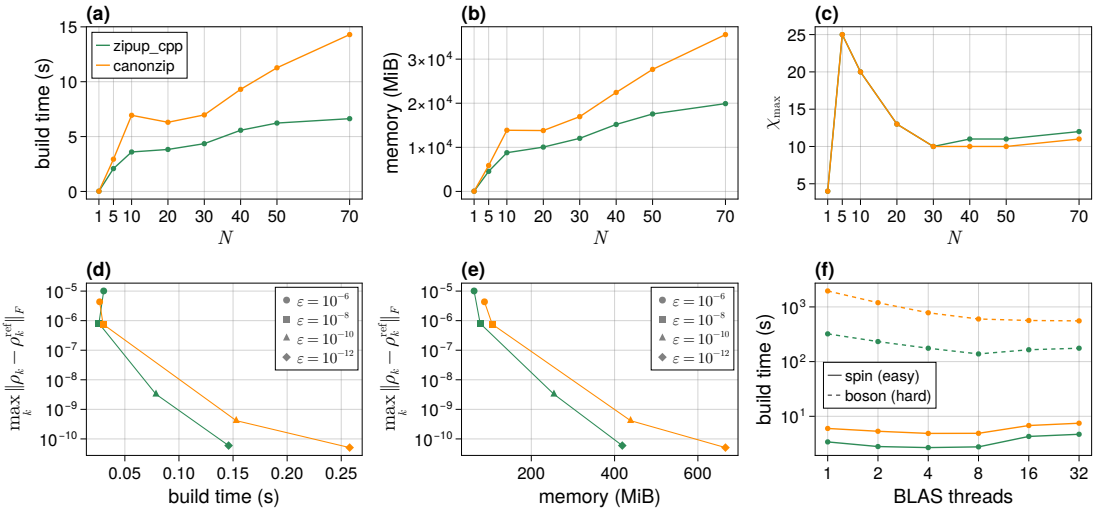


Figure 17: Comparison of the two ACE compression schedules. Panels (a)–(c) show median build time, total allocated memory, and maximum temporal bond dimension of the two compression schedules for the unpolarised central-spin scaling, benchmarked over seven samples. Panels (d)–(e) show trajectory error relative to a tightly converged reference. Panel (f) shows one benchmarked build per BLAS thread count for fixed spin (easy) and bosonic (hard) process tensors.

does not come from retaining a smaller final PT-MPO: panel (c) shows comparable bond dimensions, with `:canonzip` sometimes reaching a slightly smaller $\chi_{\max}$. The benefit of canonicalisation is clearest in panels (d) and (e). At each matched cutoff, `:canonzip` produces the smaller trajectory error, so it is the more accurate compression schedule for a fixed truncation threshold, but that accuracy is purchased with greater runtime and memory allocation.

Panel (f) isolates parallelism that can be used inside the SVDs by varying the BLAS thread count. ITensors uses LAPACK's divide-and-conquer `gesdd` routine for these decompositions, so users can accelerate sufficiently large ACE builds through a multithreaded BLAS/LAPACK backend. The "easy" case is a five-mode polarised spin bath with local Hilbert dimension 2, 200 time steps, cutoff 10^{-10}, and $\chi_{\max} = 28$. The "hard" case is an eight-mode Lorentzian bosonic bath with local dimension 5, 60 time steps, cutoff 10^{-8}, and $\chi_{\max} \simeq 50$. Its larger mode indices and temporal bonds produce much larger SVD matrices, making the build more expensive while exposing enough dense linear-algebra work for effective BLAS parallelism. For the smaller spin problem, the SVD matrices are too small to pay for the overhead of thread-startup, scheduling, and synchronization, so additional BLAS threads provide little benefit and can even increase runtime. In the bosonic case, large SVDs dominate the build and their BLAS kernels contain sufficient parallel work to produce substantial speedup.

The relative merits of canonicalising before truncation and truncating during the forward zip-up sweep were raised in Referee Report 1 and the accompanying author response in the SciPost submission thread [71]. The referee argued that canonicalising before truncation makes each SVD locally optimal, while the authors defended forward truncation as a substantially faster approximation that remains accurate in practice. Our results in Fig. 17 indeed show a trade-off rather than a universally superior schedule: `:canonzip` is more accurate at a fixed cutoff, whereas `:zipup_cpp` is faster and uses less memory. A tighter zip-up cutoff can recover the required accuracy, so the appropriate choice depends on the available memory and target error.

6.4 Benchmark against the original C++ ACE toolkit

We finally compare `ProcessTensors.jl` with the original C++ ACE toolkit of Cygorek and Gauger [22], following the same sequential ACE algorithm and contraction procedure with `:zipup_cpp` matching its compression schedule.[4] C++ ACE was compiled with g++ 13.3.0 and `-O3`, using Eigen 3.4.0 with OpenBLAS 0.3.30 and LAPACKE support, so its `JacobiSVD` interface calls LAPACK's `gesvd`. Julia 1.12.7 uses OpenBLAS 0.3.29 with ITensors' LAPACK divide-and-conquer SVD, `gesdd`. Both implementations use one application thread and one BLAS/OpenMP thread, with independent cases pinned to separate CPUs. Julia timings cover the warmed builder call; C++ timings cover the executable with propagation disabled, including setup and PT-file output. Full software versions, compiler flags, and run parameters are recorded with the benchmark scripts in the `benchmark/` directory of the main repository.

The benchmark covers fully polarised, partially polarised, and unpolarised central-spin baths, together with a Lorentzian spin–boson bath,

$$J(\omega) = \frac{C}{\pi} \frac{\gamma}{(\omega - \omega_c)^2 + \gamma^2}, \qquad \omega_c = \Omega, \quad \gamma = 0.1\Omega, \tag{98}$$

with $C/\Omega^2 = 0.2$, $M = 5$, $\Omega T = 8$, $\Omega \Delta t = 0.1$, and $k_B T/\Omega \in \{0, 0.5, 1, 3\}$. Both implementations use the same timestep, final time, bath discretisation, relative cutoff, second-order propagation choice, and bath realisation. For the partially polarised and unpolarised central-spin cases, the bath-spin orientations generated by C++ are read back and supplied directly to our Julia constructors.

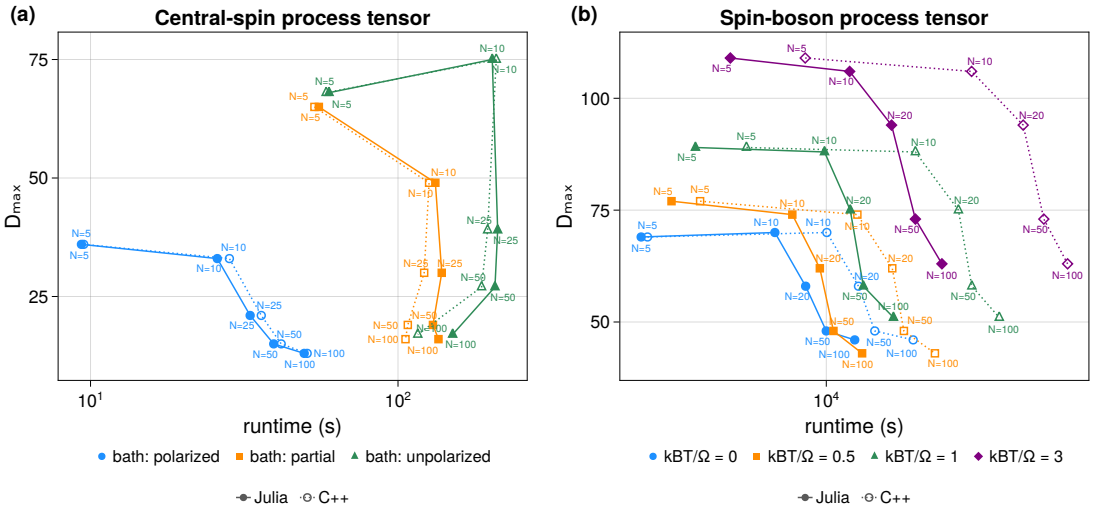


Figure 18: Runtime and maximum PT-MPO bond dimension for (a) central-spin baths at three polarisations and (b) Lorentzian spin–boson baths at four temperatures. Filled symbols denote Julia and open symbols denote C++; labels indicate the number N of bath modes.

Figure 18 compares construction time and $D_{\max}$ as the number of bath modes is increased for three central-spin polarisations (Figure 18(a)) and four spin–boson temperatures (Figure 18(b)). Across both model families, matched Julia and C++ points have the same $D_{\max}$, showing that the two implementations retain the same maximum bond dimension under the matched cutoff and propagation settings. The central-spin runtimes agree closely, with C++ slightly faster for the larger partially polarised and unpolarised baths. For the spin–boson

[4] https://github.com/mcygorek/ACE

model, Julia remains faster, particularly for cases involving larger intermediate tensors. The much closer agreement after enabling LAPACKE in C++ suggests that the numerical SVD backend accounted for much of the earlier runtime difference. The remaining difference is plausibly associated with the use of `gesvd` in C++ and `gesdd` in Julia; a comparison using the same driver would help isolate this contribution. These results therefore reflect the performance of the two implementations with their respective OpenBLAS configurations, rather than an algorithmic speedup over sequential ACE. Other numerical libraries, such as MKL on either AMD or Intel hardware, may change the relative timings.

Together these benchmarks establish the hierarchy of controls that should be reported in practical ACE calculations: the timestep and Trotter order determine propagation error; the bath grid determines which environmental correlations are represented; the SVD cutoff and `maxdim` determine how much of that representation is retained; and the compression schedule determines the intermediate cost.

7 Extensibility of `ProcessTensors.jl` and its roadmap

So far, we have demonstrated a broad range of workflows already available in `ProcessTensors.jl`, from the construction and compression of process tensors to their interrogation with instruments and testers. Nevertheless, the present implementation should be viewed as a foundation rather than a complete endpoint. A central motivation for developing `ProcessTensors.jl` as an open-source package is to provide an interface that can continue to grow with new process tensor algorithms, new classes of environments, and new operational tools, while retaining a common ITensor-based language. We have therefore designed the present API with room for extensions that may be contributed and refined by the wider community. Below we outline several directions that we consider particularly natural continuations of the current architecture, both for broadening the physical problems that can be represented and for improving the numerical construction of PT–MPOs.

7.1 Extending baths and measurement interfaces

One of the most important missing environment classes in the current implementation is a fermionic bath. Fermionic reservoirs arise naturally in quantum-transport scenarios, where a small localized electronic system is coupled to extended electronic leads [72–77]. A paradigmatic example is the resonant-level model, which has already been treated directly within ACE using the localized fermionic level as the system and the discretized lead modes as independent environmental degrees of freedom [20]. The ITensor ecosystem already provides `Fermion` and `Electron` site types together with fermionic creation and annihilation operators, fermion-parity quantum numbers, and Jordan–Wigner string operators required to represent anticommutation relations in tensor-network calculations. A natural extension of `ProcessTensors.jl` would therefore be to introduce corresponding fermionic Hilbert-space sites and to carry their parity structure consistently through the existing Hilbert-to-Liouville mapping. In Liouville space, left and right multiplication by fermionic operators must preserve the appropriate parity signs on the ket and bra sectors; once these superoperators are defined, they can enter the same Liouvillian and propagator routines presently used by bosonic and spin degrees of freedom. A `FermionicMode` and corresponding fermionic bath container could then expose this construction through the same bath-mode interface used elsewhere in the package, allowing the resulting microscopic propagators to be compressed into PT–MPOs without requiring a separate process tensor representation.

The instrumentation layer can likewise be made more operational. The current instrument primitives are sufficient to compose state preparations, observable measurements, projections,

and more general maps, and therefore already allow many POVM-based protocols to be represented. However, POVMs are not yet exposed as a first-class user abstraction that groups a complete set of measurement outcomes into a single operational object. A dedicated POVM utility could provide a collection of positive effects $\{E_a\}$, verify the completeness condition $\sum_a E_a = \mathbb{I}$ when appropriate, associate each outcome with its corresponding single-leg readout, and, when a post-measurement state update is required, connect the effects to a compatible quantum instrument. Such an interface would make experimentally motivated measurement protocols considerably easier to express than assembling each outcome manually from lower-level preparation and measurement primitives, while remaining naturally compatible with the existing `AbstractInstrument` hierarchy and `InstrumentSeq` schedules.

A further extension concerns dissipative bath modes. At present, the ACE path constructs each microscopic mode influence through a closed Hilbert-space unitary channel, even though the underlying builder infrastructure already contains a Liouville-space propagator path. This makes mode-local Markovian dissipation a particularly direct addition. Bath modes could be supplied with jump operators and rates, in the same spirit as the dissipative system objects already used elsewhere in the package. A prospective user interface could add the existing `jump_ops` convention directly to a bath-mode constructor:

Julia

```
loss = OpSum()
loss += kappa, "A", 1

mode = bosonic_mode(
    bath_sites_L,
    H_E,
    rho_E0_L;
    coupling=H_SE,
    jump_ops=[loss],
)
```

Listing 24: Prospective interface to define a lossy bosonic bath mode.

The bath-mode propagator would then determine whether a mode is closed and unitary or whether its timestep map must instead be generated from a Liouvillian containing the Hamiltonian, system–mode coupling, and mode-local jump operators. Importantly, this changes the microscopic mode propagator rather than the ACE compression itself: once the resulting single-mode influence is represented as a PT–MPO, the existing joining and compression routines can operate on it unchanged. This separation provides a straightforward route to damped cavity modes, lossy auxiliary modes, or locally thermalised environmental degrees of freedom without introducing a second compression formalism.

7.2 Scalable ACE contraction and custom bath modes

The present ACE backend combines independent mode influences sequentially, following the construction used in the original ACE toolkit [22]. A particularly promising extension is the treelike contraction scheme introduced for ACE in Ref. [21], where partial PT–MPOs are combined hierarchically rather than adding every new mode to one continuously growing process tensor. The published construction shows that this reordering can substantially reduce the cost of the intermediate contractions, while preselection and additional compression sweeps can control the large temporary bond spaces that appear when two already-compressed PT–MPOs are joined. From the perspective of the present API, this is naturally a new contraction strategy rather than a new process tensor type:

Julia

```julia
pt_tree = build_process_tensor(
    system;
    method=ACE(
        cutoff=1e-10,
        compression=:canonzip,
        contraction=:tree,
        parallel=true,
        # Other ACE kwargs
    ),
    environment=environment,
    dt=dt,
    nsteps=nsteps,
)
```

Listing 25: Prospective tree-like ACE construction through the existing builder interface.

The option `contraction=:sequential` would retain the current behaviour. Additional keyword arguments could expose treelike-specific choices such as preselection and the number of compression sweeps, without prescribing a particular internal implementation to future contributors. This keeps the public interface compact while allowing the ACE backend to incorporate improvements developed in the broader literature.

Treelike contraction also creates a natural opportunity for parallel execution. Independent pairs of partial PT–MPOs within the same tree level can be constructed and compressed concurrently before the next level is assembled. A future implementation could therefore parallelise independent branches of the contraction tree while retaining the same user-facing builder. The precise scheduling strategy should remain an implementation choice—in particular, it must balance task-level parallelism against threaded linear-algebra kernels and the memory cost of holding several intermediate PT–MPOs simultaneously—but the treelike organization provides the required independence without any change to the mathematical object returned to the user.

ACE can also be extended from static to explicitly time-dependent bath modes. The current single-mode construction reuses a timestep propagator when the bath Hamiltonian is time independent. Allowing the Hamiltonian of a bath mode, or its coupling to the system, to depend on time would instead let the builder generate a mode propagator appropriate to each interval $[t_k, t_{k+1}]$. At the API level, this could be expressed directly through the bath-mode Hamiltonian or coupling object, for example by permitting callable or otherwise time-dependent operator coefficients. A driven spin mode and its process tensor could then be constructed through a prospective interface such as in Listing 26:

Julia

```julia
function H_E(t)
    H = OpSum()
    H += omega_E / 2, "Sz", 1
    H += drive * cos(nu * t), "Sx", 1
    return H
end

driven_mode = spin_mode(bath_sites_L, H_E, rho_E0_L; coupling=H_SE)
```

Listing 26: Prospective ACE construction for a bath mode with a time-dependent Hamiltonian.

The ACE builder would then read the mode at the requested time and construct the corresponding microscopic timestep map before carrying out the usual PT–MPO combination and compression. In this way driven auxiliary modes, modulated couplings, and other non-stationary environments could be introduced without changing the representation of the final process tensor.

More generally, the ACE backend provides a natural route toward user-defined baths and bath modes. The essential requirement of ACE is not that every environmental degree of freedom belongs to one of a small number of predefined physical classes, but that the influence of each independent microscopic component can be supplied as a compatible system–mode timestep map. Consequently, a lightweight interface for custom subtypes of `AbstractBathMode` could allow advanced users to define their own local Hilbert space, initial state, generator, and system–mode coupling while reusing the same ACE machinery for PT–MPO assembly. The existing `BosonicMode` and `SpinMode` types would then remain convenient high-level constructors, while fermionic, multilevel, anharmonic, driven, or application-specific modes could enter through the same backend. Such an extension would preserve the pedagogical simplicity of the standard API while making `ProcessTensors.jl` a substantially more general platform for constructing process tensors from microscopic models.

7.3 Support for Gaussian baths using Feynman–Vernon approaches

The ACE construction used in the present implementation is deliberately microscopic: a bath is resolved into individual modes, the influence of each mode is constructed explicitly, and the resulting process tensors are combined and compressed. This flexibility is valuable for structured, non-Gaussian, or otherwise mode-resolved environments, but a continuum bath must first be represented by a finite collection of modes, introducing an additional bath-discretisation parameter whose convergence must be checked [78]. Fortunately, for stationary Gaussian environments there is a complementary route. If the bath consists of non-interacting modes and is initially in a stationary thermal Gaussian state, its influence can be integrated out analytically at the level of the Feynman–Vernon influence functional [79, 80]. The bath may then be specified directly by a spectral density $J(\omega)$ and temperature rather than by an explicit list of microscopic modes. This removes the additional error associated with discretising a continuum spectrum into finitely many bath particles, while retaining the non-Markovian temporal correlations of the bath. The remaining numerical approximations are instead controlled by the time step, numerical quadrature, any imposed memory cutoff, and tensor-network truncation. For Gaussian baths whose temporal correlations remain compressible, this can provide a particularly compact and accurate representation of the environment.

For a thermal Gaussian bath linearly coupled to the system,

$$H_{SB} = S \otimes B, \tag{99}$$

the spectral density and temperature determine the bath correlation function $C(\tau) = \langle B(\tau)B(0)\rangle$. This function describes how fluctuations of the bath at one time are correlated with those at another time, and contains the information needed to construct its influence on the system.

To describe this influence on a discrete time grid, we integrate $C(t-t')$ over pairs of time intervals. For two distinct intervals, $j < k$, this gives

$$\eta_{k-j} = \int_{t_k}^{t_{k+1}} \mathrm{d}t \int_{t_j}^{t_{j+1}} \mathrm{d}t'\, C(t-t'). \tag{100}$$

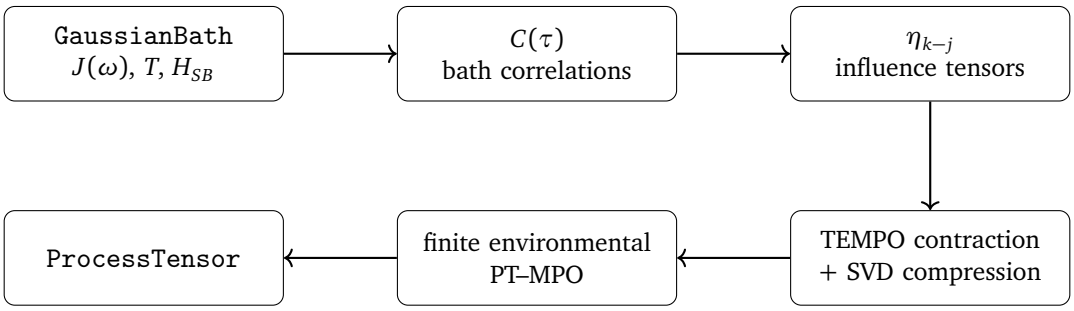


Figure 19: Proposed PT-TEMPO construction in the present package. A stationary Gaussian bath is specified directly through its spectral density, temperature, and system–bath coupling. The corresponding two-time correlations generate the discretised Feynman–Vernon influence tensors, which are contracted and compressed into a finite environmental PT–MPO. Free-system propagation is then embedded using the same convention as the existing process tensor builders, producing the standard `ProcessTensor` object used by the rest of the package.

For the same interval, $j = k$, the inner integral instead ends at t, retaining the time ordering $t' \leq t$. On a uniform grid, stationarity makes these coefficients depend only on the separation $k-j$. Each coefficient determines an influence tensor connecting the system's ket and bra histories in the corresponding intervals. Together, these tensors form the discretised Feynman–Vernon influence functional: a tensor network describing how the bath connects the system's evolution at different times. PT-TEMPO contracts and compresses this network into a PT–MPO, keeping the system legs available for subsequent insertion of propagators and quantum instruments. The explicit influence tensors and their contraction are described in the original TEMPO and PT-TEMPO references. TEMPO rewrites this non-local temporal object as a tensor network and repeatedly applies singular-value decompositions to retain only the relevant temporal correlations [81]. PT-TEMPO takes the same construction one step further: rather than contracting the network for one chosen reduced-state trajectory, the system input and output legs are retained, so that the compressed influence functional becomes a reusable finite process tensor MPO [82]. Once constructed, this object can be interrogated with different controls, instruments, and multitime protocols without rebuilding the bath influence, as demonstrated in existing PT-TEMPO software such as OQuPy [83].

The present source layout already provides most of the interfaces required for such a builder. We envisage a `GaussianBath <: AbstractBath` in the existing environment module, storing three physical ingredients:

- **Temperature** T: the thermal scale of the stationary Gaussian state.
- **Spectral density** $J(\omega)$: an Ohmic or Lorentzian form, or a user-supplied $J(\omega)$.
- **Linear coupling** H_{SB}: an operator sum validated against the Gaussian influence-functional construction.

Unlike the existing spin and bosonic bath containers, this object would not contain a finite collection of individual bath modes: the continuum has already been encoded in $J(\omega)$. The coupling would then be separated internally into its system and bath components. A PT-TEMPO method would sit beside the existing dense and ACE constructors as an `AbstractPTBuilder`, so that a prospective interface could assemble the bath and the process tensor together, as in Listing 27:

Julia

```julia
bath = gaussian_bath(J, T; coupling=H_SB)

process_tensor = build_process_tensor(
    system;
    environment=bath,
    method=PTTEMPO(
        #kwargs...
    ),
    dt=dt,
    nsteps=nsteps,
)
```

Listing 27: Prospective Gaussian-bath constructor and PT-TEMPO process tensor builder.

The implementation itself would reside in a new builder file beside the dense and ACE backends. Internal helpers in this file would evaluate the bath correlations from $J(\omega)$ and T, construct the discretised influence tensors, and perform the TEMPO contractions and SVD compression. A corresponding dispatch would return the same finite process tensor representation used throughout the package. The existing ITensor infrastructure supplies the MPO representation and bond compression; the only additional numerical dependency expected at this level is an adaptive quadrature library, such as `QuadGK.jl`, for the spectral and influence-functional integrals. Free-system propagation can then be embedded after the environmental PT–MPO has been constructed, and may be chosen trivial when only the bath influence is required.

The same Gaussian-bath abstraction could later accept fermionic hybridisation functions in place of bosonic correlation functions. An ITensor-native route would order fermionic events along the temporal contour and encode their exchange signs through Jordan–Wigner parity strings, or equivalent parity sectors, producing an ordinary complex-valued but parity-aware influence MPO. Fermionic influence-functional MPS calculations using closely related superfermion and Slater-determinant machinery have already been implemented for quench dynamics of the Anderson impurity model [84]. Grassmann TEMPO provides an alternative formulation [16, 85], but incorporating its Grassmann tensor algebra would require substantial changes to the present backend.

Our immediate goal would be the standard finite-time PT-TEMPO setting: stationary thermal bosonic Gaussian baths with non-interacting modes, linear system–bath coupling, user-defined or built-in spectral densities, and controlled temporal SVD compression. More advanced influence-functional contractions can be incorporated only after this reference implementation is established. In particular, divide-and-conquer constructions can reduce the number of SVDs by exploiting self-similarity of Gaussian influence networks [12], while time-translationally invariant finite-time constructions can build bosonic and fermionic matrix-product representations of the influence functional with substantially fewer tensor-network multiplications [13, 86]. The latter tensors still grow with the requested time window, and exponential kernel fitting introduces additional convergence parameters, so fewer multiplications do not imply time-independent total cost. These algorithms provide natural future optimisation routes, but are not prerequisites for adding a self-contained PT-TEMPO backend to this package.

7.4 Chain-mapped environments and temporal tensor-network contractions

A complementary long-term direction is to construct process tensors from an explicit tensor-network simulation of the joint system–environment dynamics. Orthogonal-polynomial transformations map a system linearly coupled to a continuum of bosonic or fermionic modes onto a semi-infinite nearest-neighbour chain whose onsite energies, hopping amplitudes, and system–chain coupling are fixed by the bath spectral density [87, 88]. This mapping is exact before numerical truncation and places the environment in a geometry naturally suited to MPS/MPO methods. Finite-temperature extensions such as T-TEDOPA additionally permit thermal environments to be represented through temperature-dependent chain parameters and pure-state propagation [89]. For finite simulations the chain length and, for bosons, the local Hilbert spaces must still be truncated, but rigorous error bounds are available for these approximations [90]. Compared with ACE, this route does not require the bath to remain a collection of mutually independent modes during the simulation; compared with PT-TEMPO, it does not rely on evaluating the Gaussian Feynman–Vernon influence functional analytically. More interestingly, this formalism can accept initially correlated bath as an initial condition (although the bath modes still don't talk to each other). It therefore offers a potentially broader route to strongly coupled and structured environments, at the price of explicitly propagating a many-body bath.

The corresponding process tensor can be understood by viewing the discretised joint propagation as a two-dimensional space–time tensor network [37]. Standard TEDOPA contracts this network in the time direction and compresses the evolving system–environment state spatially. An alternative is to contract bath sites successively in the spatial direction, leaving the system legs open at every timestep. The resulting object is a temporal MPS/MPO encoding the influence of the eliminated environment, and can therefore be interpreted directly as an influence functional or finite process tensor. Iterative constructions of this kind have already been demonstrated for general quantum environments by Ye and Chan [25], while the relation between transverse tensor-network contraction, influence functionals, and process tensors has been reviewed recently in Ref. [26]. In favourable regimes, singular-value compression along the temporal bonds can make this representation substantially more compact than retaining the full propagated environment state.

For `ProcessTensors.jl`, a future implementation could therefore follow the schematic pipeline

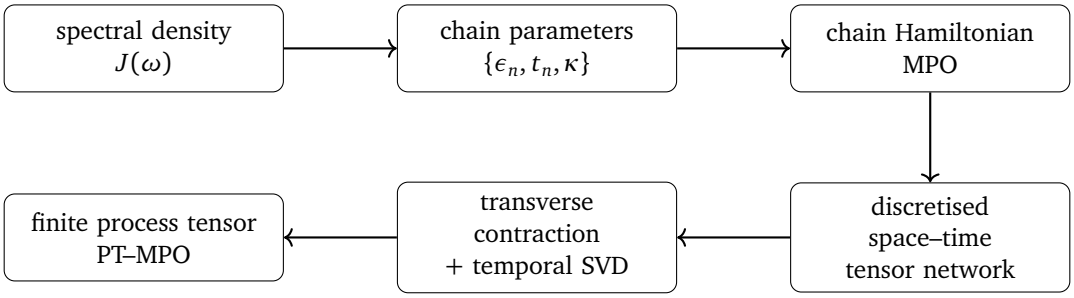


Figure 20: Prospective chain-mapping route from a continuum spectral density to a finite temporal PT–MPO.

The existing Julia package `MPSDynamics.jl` already provides mature T-TEDOPA chain mappings and MPS dynamics for bosonic and fermionic environments [91], and therefore provides a useful reference implementation and benchmark. We would nevertheless favour a lightweight chain-mapping layer compatible with the current ITensors backend rather than make a second tensor-network framework a required dependency.

This direction is more ambitious than the ACE and PT-TEMPO extensions discussed above. Its efficiency is controlled by the temporal entanglement generated during transverse contraction: the final influence functional may be compact even when intermediate tensors become substantially more entangled, and generic ergodic environments can require rapidly growing temporal bond dimensions [25, 26]. Nevertheless, if this compression remains favourable, chain mapping would provide a powerful third route to finite PT–MPOs: ACE compresses explicitly separated bath modes, PT-TEMPO compresses analytically obtained Gaussian influence functions, while transverse chain contraction would construct the bath influence directly from an explicit many-body environment evolution.

7.5 Quantum memory witness using semidefinite programs

Non-Markovianity tells us that the environment retains information about the system's history, but not whether this information requires a *quantum* memory. Much like an entanglement witness distinguishes entangled states from separable ones, a quantum-memory witness seeks to rule out a specified class of processes explainable by classical memory. Giarmatzi and Costa showed how separability criteria on an appropriate process representation can reveal quantum memory [92]. More recently, Sander *et al.* developed witness searches based on history–future responses to interventions across a causal break [93]. These results motivate extending `ProcessTensors.jl` from constructing and interrogating non-Markovian processes to certifying the quantum character of their memory. Following the witness construction of Sander *et al.*, let Λ be a linear map such that $\Lambda(\sigma) \succeq 0$ for every process operator σ in the specified classical-memory ansatz. For a suitably defined history–future Choi operator Υ_{HF}, a witness search can then take the semidefinite-program form

$$\begin{aligned} \min_{P,Q} \quad & \mathrm{Tr}[W\Upsilon_{HF}], \\ & W = P + \Lambda^\dagger(Q), \qquad P,Q \succeq 0, \\ & \mathrm{Tr}(P+Q) = 1, \end{aligned} \tag{101}$$

where $\Lambda^\dagger$ is the adjoint map. By construction, $\mathrm{Tr}(W\sigma) \geq 0$ for every σ in the chosen classical-memory set, so a negative optimum certifies quantum memory relative to that ansatz. Partial transposition, $\Lambda = T_F$, is a familiar special case: it yields the decomposable form $W = P + Q^{T_F}$ used in entanglement-witness searches [92]. Julia's `JuMP.jl` modelling interface and semidefinite solvers such as `Clarabel.jl` offer a direct numerical route [94, 95].

A first implementation could use the existing `evaluate_process` interface to contract unwanted temporal legs with specified instruments or deterministic closures, while retaining only the history and future legs of interest. The resulting partially contracted dense `ITensor` can be reshuffled into a reduced Choi operator for the SDP in Eq. (101). Restricting the search to a small temporal window is important: although the full process is stored as a compressed PT–MPO, the dimension of its dense Choi representation grows exponentially with the number of retained intervention times. The selected closures must preserve the intended operational interpretation of the history–future division; a witness on this reduced object certifies the corresponding reduced process, while an inconclusive result does not rule out quantum memory in the full process.

A more ambitious extension would optimise witnesses directly against the Choi PT–MPO, avoiding its dense reconstruction. Inspired by locally purified matrix-product representations and matrix-product quantum channels [96–98], one could parameterise the positive parts of a witness associated with the chosen map Λ using auxiliary MPOs X and Y,

$$W = X^\dagger X + \Lambda^\dagger\big(Y^\dagger Y\big), \qquad \mathrm{Tr}(X^\dagger X + Y^\dagger Y) = 1. \tag{102}$$

Provided Λ satisfies the classical-memory positivity condition above, this construction preserves the corresponding witness constraint throughout a variational search, while $\mathrm{Tr}(W\Upsilon_{HF})$ can be evaluated by tensor-network contractions. The main challenges are choosing sufficiently expressive auxiliary bond dimensions and optimising a non-convex MPO ansatz; moreover, the chosen map and finite-bond ansatz need not detect every form of quantum memory. The feasibility of this approach additionally depends on whether the chosen positive map Λ admits an efficient tensor-network representation. Local maps such as partial transposition can be represented efficiently, but a generic nonlocal map may require increasing the temporal bond dimensions. Nevertheless, a reliably negative expectation value would still provide a valid certificate, making this a promising tensor-network-native counterpart to the small-window dense SDP.

8 Conclusion

We have introduced `ProcessTensors.jl`, an ITensor-native Julia framework for constructing, compressing, and interrogating finite-time process tensors. A central emphasis of this work has been to make the underlying theory accessible to readers who may be unfamiliar with the process tensor formalism. To this end, we have developed the theoretical concepts progressively, using tensor diagrams to provide pictorial interpretations of the mathematical expressions involved in Liouville-space dynamics, quantum operations, and process tensor construction. Particular care has been taken to maintain a close correspondence between these equations, their graphical representations, and the accompanying ITensor implementations, including the index conventions used to assemble and contract the resulting tensor-network objects. We hope that this approach lowers the barrier to understanding process tensors and allows new users to develop the necessary intuition while learning to use the package.

The present implementation provides dense and sequential ACE algorithms for constructing process tensors from microscopic environments, together with an instrument and tester framework for studying their operational properties. Through several examples, we have demonstrated how a single constructed PT-MPO can be reused to investigate reduced dynamics, multi-time correlations, measurement protocols, and memory-assisted control. We have also examined the numerical accuracy, temporal compressibility, and computational costs of the ACE implementation, including a benchmark against the original C++ ACE toolkit under documented numerical settings. These results establish the capabilities and practical limitations of the current implementation, while demonstrating how the Julia and ITensor ecosystems can be used to construct and manipulate non-Markovian quantum processes.

The theory and computational methods surrounding process tensors continue to develop rapidly, and we do not intend `ProcessTensors.jl` to be an exhaustive implementation of every available approach. Rather, we envision it as an evolving framework whose continued relevance will depend on contributions from the wider research community. With this in mind, particular attention has been given to a modular architecture in which new environment models, construction algorithms, and operational tools can be incorporated while retaining a common process tensor representation and its existing interfaces. We have outlined several promising directions, from improved ACE compression and alternative bath constructions to semidefinite and tensor-network approaches for witnessing quantum memory. Through its open-source implementation, examples, and documentation, we hope that `ProcessTensors.jl` will serve both as an accessible introduction to process-tensor methods and as a collaborative platform for their continued development and application in non-Markovian quantum dynamics.

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Disclosure of generative AI use The authors disclose that large language models accessed through coding assistants were used to draft `TikZ` figures, to generate example code listings that were subsequently verified by hand, and to assist in the construction of `ProcessTensors.jl`. All AI-generated source, including figure code, listing code, and package code, was examined thoroughly by the authors and tested against the intended mathematical and numerical behaviour. The authors take full responsibility for the contents of the manuscript and of the software.

Author contributions Dario and Modiji, do we need this? Contributions should be succinctly described in a single short paragraph, using author initials.

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Code and data availability The development repository is available at https://github.com/Gauthameshwar/ProcessTensors.jl and the documentation at https://gauthameshwar.github.io/ProcessTensors.jl/stable/. **[TODO: insert release DOI/Zenodo or SciPost codebase DOI at submission.]**

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A How to read the Julia listings

Code examples throughout this paper are displayed in framed Julia listings. A blue **Julia** badge in the top-right corner identifies the language. Printed program output is shown in a matching gray box with a **Terminal** badge; that text is not syntax-highlighted. Line numbers on the left are the ones referred to in the text (for example, Listing 28, lines 4–6). Syntax highlighting is chosen to make the origin of each identifier visible at a glance, rather than to reproduce a particular editor theme.

The colour roles are as follows.

- **Blue, bold:** Julia keywords such as `using`.
- **Violet, bold:** names exported by `ProcessTensors.jl`. This includes package-specific constructors and converters (`to_dm`, `liouv_sites`, `liouvillian_mpo`, `tester`, ...), tensor-network types such as `MPS`, `MPO`, and `OpSum`, and re-exported ITensorMPS inspection routines such as `siteinds`, `linkinds`, and `inner`.
- **Teal, bold:** functions that belong to `ITensors.jl` and are *not* part of the `ProcessTensors.jl` export list (`Index`, `ITensor`, `dag`, `prime`, `delta`, `op`, `apply`, ...).
- **Orange, bold:** package types and backends such as `Hilbert`, `Liouville`, `Dense`, `ACE`, `InstrumentSeq`, and `TesterSeq`.
- **Near-black:** ordinary variables.
- **Green, bold:** numeric literals.

- **Crimson:** string literals.
- **Grey italic:** comments.

Listing 28 collects these conventions in one place. Violet names are the objects a user of `ProcessTensors.jl` is expected to call; teal names are the underlying ITensor primitives on which those objects are built.

Julia

```julia
using ITensors, ProcessTensors

s = Index(2, "s")                          # ITensor index
psi = ITensor([1.0, 0.0], s)
sites = siteinds("S=1/2", 1)

rho = to_dm(MPS(sites, ["Up"]))            # ProcessTensors converter +
 ↪  MPS
H = OpSum()
H += 0.7, "Sx", 1
L = liouvillian_mpo(H, liouv_sites(sites))
```

Listing 28: Colour legend for the listings in this paper.

B Time dynamics in Liouville space

The Liouville-space MPS and MPO objects introduced in Sec. 2.4 provide a direct representation of time-dependent open-system states and generators. If $|\rho(0)\rangle\rangle$ is an initial Liouville MPS and $\mathcal{L}$ is its Liouvillian, the state at time t is

$$|\rho(t)\rangle\rangle = \exp(t\mathcal{L})|\rho(0)\rangle\rangle. \tag{B.1}$$

`ProcessTensors.jl` extends the time-evolution algorithms provided by `ITensorMPS.jl` to its typed Hilbert- and Liouville-space objects.[5] These routines are useful for time-local master equations and provide reference trajectories against which process tensor calculations can be validated.

For a small two-spin example, Liouville-space TEBD approximates the propagator over successive time steps by constructing local gates for $\exp(\Delta t\,\mathcal{L})$ from a Hilbert-space Hamiltonian and local dissipators [50, 51, 99, 100]:

Julia

```julia
using ITensors, ProcessTensors
using ITensors.Ops: Trotter

sites = siteinds("S=1/2", 2)
sites_L = liouv_sites(sites)
psi0 = MPS(sites, ["Up", "Dn"])
rho_L0 = to_liouville(to_dm(psi0); sites=sites_L)

```

[5] The gate-based MPS time-evolution interface is documented at https://docs.itensor.org/ITensorMPS/stable/tutorials/MPSTimeEvolution.html.

```
H = OpSum()
H += 1.0, "Sz", 1, "Sz", 2
H += 0.4, "Sx", 1
H += 0.4, "Sx", 2
jumps = [(0.15, "S-", 1), (0.15, "S-", 2)]

dt, T = 0.025, 0.1
rho_L_tebd = tebd(
    rho_L0, H, dt, T;
    jump_ops=jumps,
    alg=Trotter{2}(),
    maxdim=32,
    cutoff=1e-12,
)
```

Listing 29: Time evolution of a dissipative Liouville-space state with TEBD.

TDVP provides a complementary time-evolution route by acting with the Liouvillian MPO itself. The evolution time supplied to TDVP is real because the Hamiltonian contribution $-i[H,\rho]$ and all dissipative terms are already contained in $\mathcal{L}$:

Julia

```
L = liouvillian_mpo(H, sites_L; jump_ops=jumps)
rho_L_tdvp = tdvp(
    L, T, rho_L0;
    time_step=dt,
    nsite=2,
    maxdim=32,
    cutoff=1e-12,
    outputlevel=0,
)
```

Listing 30: Time evolution of the same Liouville-space state with TDVP.

The two methods expose the same physical time dynamics through different tensor-network approximations: TEBD applies compressed local gates from the second-order Strang splitting [56, 101], whereas two-site TDVP projects the Liouvillian evolution onto the MPS tangent space and integrates it [53, 54, 102]. In either case, convergence should be checked with respect to the time step, truncation cutoff, and maximum bond dimension. The resulting trajectory should also be validated by monitoring trace, Hermiticity, and positivity.

Complete discussions of Trotter order, TDVP parameters, dense validation, and time-resolved observables are available in the unitary-dynamics and dissipative-dynamics tutorials. Larger time-dependent examples, including dissipative spin chains, boundary-driven transport, and driven bosonic systems, are linked from the stable package documentation. For a more detailed overview of time evolution using MPS and MPO, refer to the review by Paeckel et al. [52].

C Sequential ACE construction

The ACE builder introduced in Sec. 3.5 constructs the environmental influence by adding one microscopic mode at a time and compressing the resulting temporal bonds. Here we describe

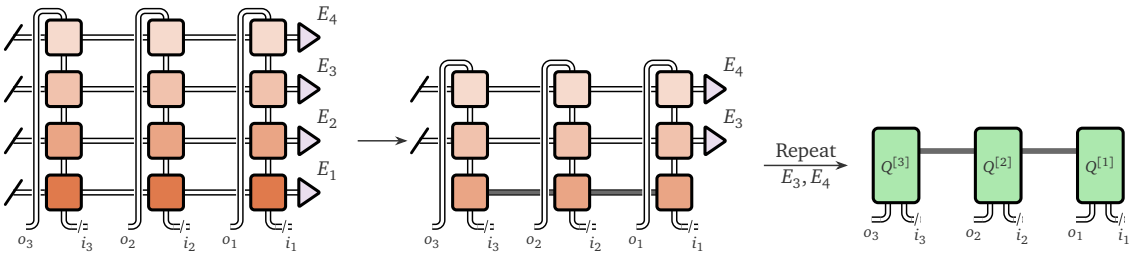


Figure 21: Sequential construction of the PT-MPO, illustrated for four bath modes and three propagation intervals, with time running from right to left. Each bath mode E_m contains an initial state and its propagator across time. The vertical legs of each bath mode contain the influence of that bath mode on the system. By merging this leg sequentially for each bath mode with SVD truncation, we can retain only the relevant influence on the system at each timestep. Repeating this over all bath modes gives the final PT-MPO. Figure adapted from Lacroix *et al.* [38].

this update in more detail, following the original ACE construction [20] and its implementation in the ACE toolkit [22], The system operations remain open during construction, so the resulting PT-MPO can subsequently be used with different instrument sequences.

Figure 21 shows the join for four modes and three propagation intervals, with time running from right to left. Each row is one mode, shaded from dark (first bath mode) to light (last bath mode). We assume an initially factorised state $\rho_S(0) \otimes \bigotimes_m \rho_{E_m}(0)$ and the generator decomposition given in Sec. 3.5: each $\mathcal{L}_m \equiv \mathcal{L}_{E_m+SE_m}$ acts on the system and mode m, with no direct interactions between modes. Joining successive bath indices, contracting the first with $|\rho_{E_m}(0)\rangle\rangle$, and closing the last with $\langle\langle I_{E_m}|$ gives a single-mode influence chain. The first step absorbs E_1 and E_2 into a single chain; the same join is then repeated for E_3 and E_4. Its internal bath index has dimensions d_m^2, where d_m is the mode Hilbert-space dimension. As we compress the modes, we lose the "double-wire" space and replace it with a compressed influence, with memory bond links taking its place. After every mode has been absorbed and the temporal bonds compressed, the network is the PT-MPO with cores $Q^{[k]}$.

Let $\mathcal{L}_{<m} = \sum_{j<m} \mathcal{L}_j$ denote the contribution of the modes already included. The asymmetric addition of mode m uses

$$e^{\Delta t(\mathcal{L}_{<m}+\mathcal{L}_m)} = e^{\Delta t \mathcal{L}_m} e^{\Delta t \mathcal{L}_{<m}} + \mathcal{O}(\Delta t^2). \tag{C.1}$$

The symmetric addition of mode m uses

$$e^{\Delta t(\mathcal{L}_{<m}+\mathcal{L}_m)} = e^{\frac{\Delta t}{2}\mathcal{L}_m} e^{\Delta t \mathcal{L}_{<m}} e^{\frac{\Delta t}{2}\mathcal{L}_m} + \mathcal{O}(\Delta t^3). \tag{C.2}$$

This decomposition does not matter if we have commuting interaction terms of each bath mode with the system. However, if the bath modes are non-commutative in nature, the latter would lead to a more accurate compression. Compression reduces the enlarged temporal bonds while we merge the two MPO chains. At a sweep step, the appropriate core and any factor carried from its neighbour are reshaped into a matrix and decomposed as

$$M = U\Sigma V^\dagger \longrightarrow U_r \Sigma_r V_r^\dagger, \qquad \sigma_j > \epsilon\sigma_1, \qquad r \le \chi_{\max}. \tag{C.3}$$

One factor becomes the updated core, while the remaining factor is absorbed into the next core in the sweep direction. Sweeping through the chain carries information from the initial state and final trace into the intermediate tensors [20]. Its dimension is determined by how well that influence can be compressed over the chosen time interval.

The two schedules compared in Sec. 6.3 differ in when this truncation is performed. The `:canonzip` schedule completes the mode join without truncation, canonicalises towards the

final time, and then performs a reverse relative-SVD sweep. The `:zipup_cpp` schedule truncates during the forward join and subsequently sweeps backward, reducing intermediate storage but choosing truncation bases before the complete joined chain has been canonicalised. The same numerical cutoff therefore need not give the same accuracy in both schedules. We refer to Sec. 6.3 for their measured accuracy and computational cost.

Driven or dissipative modes modify the first ingredient, while tree-like ACE joins partial influences hierarchically and introduces preselection and further compression sweeps [21]. These are the extension points described in Sec. 7 and can retain the same final `ProcessTensor` representation. Initially correlated modes or direct intermode interactions instead require revisiting the independent-mode decomposition used above.